International Journal of Machine Learning and Cybernetics (2026) 17:111
https://doi.org/10.1007/s13042-026-02994-w

ORIGINAL ARTICLE

Explainable artificial intelligence models for detecting suspicious bank transactions

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Received: 7 March 2025 / Accepted: 1 January 2026 / Published online: 16 February 2026
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Abstract

Detecting financial crime is a complex challenge due to evolving criminal strategies and fragmented detection systems, particularly in the areas of money laundering and fraud. While it is easy to implement, traditional rule-based approaches lack adaptability to new threats, and machine learning models, though more effective, often function as opaque "black boxes," limiting their practical use in regulated domains like banking, where interpretability and accountability are essential. This research presents a novel framework that combines intrinsic and post-hoc XAI techniques to detect suspicious bank transactions. Intrinsic methods provide model-inherent transparency, while post-hoc methods offer behavior-level explanations, enabling robust cross-verification of the model's decision logic. This dual-layered explainability supports both global understanding of the model and local interpretability for individual predictions. Experimental evaluation on synthetic datasets shows that the proposed framework achieves promising results, with F1 scores ranging from 0.54 to 0.63 for anti-money laundering (AML) detection and 0.66 to 0.78 for fraud detection, depending on the underlying model architecture. The use of explainability tools also enables the identification of key features shared across fraud and AML cases, revealing structural similarities. These insights demonstrate that a unified model can effectively capture both crime types, offering practical benefits for integrated financial crime risk management. By addressing the information silos caused by different regulatory frameworks and departmental responsibilities, this work supports the development of joint investigative systems and facilitates better communication between compliance teams. In addition, the explainable nature of the models helps satisfy regulatory scrutiny while improving investigator trust and decision support. The proposed approach contributes to the growing field of interpretable machine learning by providing a scalable, auditable, and regulator-friendly solution for detecting complex financial crimes.

Keywords Anti-money laundering · Fraud · Suspicious transactions

1 Introduction

Organised and financial crimes are among the most pressing global challenges, increasingly recognised as policy priorities by national and international bodies and corporations [1, 2]. Financial crime is described as a variety of criminal actions such as money laundering, terrorism financing, corruption, proliferation financing, sanctions violations, cyber financial crime, bribery, and different fraud types like market manipulation, securities fraud, insider trading, bank fraud, credit card fraud, check fraud, insurance fraud, payments fraud, medical fraud, corporate fraud, etc. and all other crimes threatening the assets [1, 3–7]. It has become a global problem that demands an international response [8–10]. Financial institutions now face heightened complexity in combating financial crime after the COVID-19 pandemic

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[11, 12]. Consumers gradually embraced a digital-first approach to financial interactions in the pre-pandemic era [13–15]. However, as the pandemic spread, consumers quickly embraced a digital lifestyle, further amplifying this transition.

Fraudsters and launderers are producing new and innovative techniques for committing crimes, which will obscure the origins of the funds and sometimes give apparent legitimacy to their activity [11, 16–18]. Crime tactics demonstrate the ability to quickly adapt to emerging trends, vulnerabilities, and prevention measures [5, 19, 20]. They demonstrate remarkable levels of customisation to the individual channels they operate (and leveraging cross-channel tactics) [21–23]. Existing detection processes are limited in distinguishing malicious behaviour from ordinary financial activity [1, 8, 24]. Financial investigation officers, therefore, need to continually review their products, services, and individual customers to ensure AML and counter-terrorism financing (CTF) programs, processes, and training match the changing levels of risk [25]. Organisations have a risk-based monitoring program to help employees identify suspicious transactions [26–28]. In Australia, for example, failure to track and report a suspicious transaction about AML or fraud to the Australian Transaction Reports and Analysis Centre (AUSTRAC) and Australian Financial Crimes Exchange (AFCX) is considered a breach of regulations. AUSTRAC, AFCX, and local Australian AML rules will seriously consider this situation, leading to heavy penalties [28, 29]. It is a reputation and trust issue for financial organisations as well [10, 14, 30]. Regulators across the globe are reinforcing the framework by implementing stricter rules [15, 19, 29]. All suspicious matter reports (SMRs) are reported to AUSTRAC in Australia. Similarly, all fraud cases are reported to AFCX for further action. To date, there have been several stages in identifying suspicious transactions [25, 31]. (i) Applying human intelligence, (ii) applying statistics, and (iii) applying machine learning models.

Researchers have built various models for financial crime to identify suspicious transactions [32, 33]. However, these models need to thoroughly explain the justification behind how and why the transactions have been identified as suspicious [34–36]. Scientists have used various detection and prevention methods based on artificial intelligence (AI) methods, non-AI methods, and the application of XAI techniques [37, 38]. Hence, the necessity of XAI for detecting financial crimes has led to new paths for innovations and implementation [39].

Although machine learning techniques have been deployed to improve detection accuracy, many of these models function as "black boxes," offering little transparency into their decision-making processes. This limitation is particularly problematic in the financial sector, where regulatory scrutiny demands clear, justifiable explanations for automated decisions. The statistical viewpoint in explainable machine learning emphasizes quantifying and validating feature attributions through rigorous, model-agnostic methods. Bracke et al. [40] employ Shapley-based Quantitative Input Influence to assess feature contributions and test robustness across subgroups. Joseph et al. [41] introduces Shapley regressions to enable formal inference, including hypothesis testing, from ML explanations. Calzarossa et al. [42] proposed a statistical framework to assess explainability based on robustness, where explanation quality is evaluated by measuring the model's sensitivity to small changes in input data. Together, these frameworks promote explainability grounded in empirical reliability and statistical soundness [41, 42]. A significant insight from the research is that fraud and AML events often overlap. Traditional financial crime detection systems tend to treat fraud and money laundering (AML) as separate entities due to differing regulatory mandates, resulting in fragmented or siloed investigative approaches and missed opportunities to uncover interconnected activities.

This study addresses these challenges by developing a unified framework capable of simultaneously detecting fraud and AML activities while ensuring interpretability through the use of both intrinsic and post-hoc XAI techniques. By identifying shared patterns and common features between fraud and AML cases, the model not only enhances detection accuracy but also facilitates a deeper understanding of how individual features contribute to model predictions.

Testing with real data is challenging because of the difficulty associated with sensitivity and strict privacy regulations.. The data cannot be shared across different organizations. Various sources have been explored for public, real, and synthetic datasets. However, it was observed that there is no data available that has both fraud and AML data, i.e., the dataset has either fraud or AML but not both. A synthetic dataset was generated to facilitate experimentation on a dataset containing both AML and fraud transactions. The contribution of this paper can be summarised as follows:

- In the first phase, the research will focus on identifying suspicious transactions using Supervised Machine learning algorithms such as random forest (RF), support vector machines (SVM), and gradient boosting (GB). This phase involves a separate analysis of suspicious transactions across three categories: (1) money laundering alerts, (2) fraud alerts, and (3) a combination of money laundering and fraud alerts.
- In the second phase, the outcomes of the first phase are then interpreted and compared for a better understanding using posthoc XAI techniques. The potential for a

combined risk control framework for fraud and money laundering will be studied. This will help the regulators by providing a combined framework that may need to be built to reduce AML and fraud rates.

A Python script was developed to create a synthetic dataset encompassing various financial transaction records. The dataset supports classification tasks for both fraud and AML labels, incorporating diverse and realistic feature values. This dataset includes many features to mimic real-world financial transactions, enhanced using the Faker library for added realism. Features for fraud and AML include categories such as transactional, account, customer behavior, technological footprints, fraud typology, etc.

Specifically, our study presents the following novel contributions:

- We conduct a comprehensive comparison of multiple XAI methods—including SHAP, LIME, decision trees, logistic regression, and Shapley-Lorenz values—across two key domains in financial crime detection: fraud and AML. Most existing works focus on one domain or a single explanation method, whereas our study provides a cross-domain, cross-method benchmark.
- We incorporate both the interpretational and statistical viewpoints in our evaluation framework, aligning with recent advancements in explainability research [40–42], and we introduce normalized explanation techniques (Shapley-Lorenz values) that improve the fairness, clarity, and comparability of feature importance.
- Our analysis includes a practical interpretability assessment of how explanations vary by technique and application, offering insights directly applicable to real-world use cases where regulatory transparency, auditability, and safe machine learning adoption are critical.

These elements collectively advance the state of the art in XAI for financial crime detection by proposing a unified, comparative, and robustness-aware explainability framework. These contributions serve both researchers and practitioners by offering a practical, statistically grounded roadmap for deploying explainable machine learning in high-stakes regulatory environments.

The rest of this paper is organised as follows: Sect. 2 briefly presents the related work. Section 3 describes the proposed methodology. Section 4 presents experimental results and discussions. Section 5 presents the conclusions and directions for future work.

2 Related work

Financial institutions are on the front lines of the battle against financial crime and, as such, must devote substantial human and technological resources to mitigation efforts. This section provides a structured review of detection and prevention strategies, classified into three categories: non-AI approaches, AI-based techniques, and the application of XAI methodologies [43–46].

2.1 Non-AI-based methods

Traditional non-AI-based methods for detecting suspicious banking transactions primarily rely on fixed, rule-based systems that are not data-driven. Although these approaches were widely used and effective in earlier contexts, they have become increasingly inadequate in addressing the complexities of modern financial crime. Nonetheless, for completeness, several representative techniques are briefly reviewed in this section.

2.1.1 Anomaly detection

In simple terms, an anomaly is an observation that deviates so much from the other observations as to arouse suspicion that it was generated by a different mechanism [13, 47]. In the financial crime area, anomalies are identified using data mining techniques. Broadly, there are three types of anomalies, as outlined below [48]. They are point, contextual, and collective anomalies. Some of the challenges in anomaly detection are [38, 49, 50]: Defining a region or norm that distinguishes what is normal and what is abnormal behaviour. Often, malicious activity causes anomalies, though they are genuine transactions, leading to unnecessary efforts by the analysts and regulators as part of the investigation.

2.1.2 Risk rating

Risks for financial institutions arise from diverse factors, including vulnerabilities to fraud and financial crime inherent in automation and digitisation, massive growth in transaction volumes, and the greater integration of financial systems within countries and internationally [1, 20]. A risk-based strategy acknowledges that the risks associated with money laundering/terrorism financing vary depending on the organisation, consumer, and transactions. It acknowledges that the corresponding company is in the best position to comprehend the specific money laundering/terrorism financing dangers, taking into consideration their [29]:

- Customer types (e.g., natural persons, companies, etc.).
- Products or services offered.

- Geographical areas in which the business operates.
- Delivery channels (e.g., over-the-counter, online).

No matter how careful organisations are in assessing and categorising the risk of their customers, the risk with individuals and organisations is very agile. This means a customer who was categorised as low may eventually turn out to be high. So, the risk rating is a mitigation process for financial crime and aids in controlling fraud and money laundering.

2.1.3 Rule-based

Historically, suspicious transaction alert systems based on a set of fixed threshold rules [35, 51] were used to flag suspicious transactions that then underwent further manual control. Such systems are still being used today in most financial institutions. In simple terms, the business rules are written against which the transactional data is run to trace suspicious transactions. The limitation is that the business rules or scenarios are written based on historical alerts (originally outlined as suspicious transactions) that were identified as money laundering cases. A business rule that predicts a new trend or scenario will not exist to trace the suspicious transaction, i.e., it cannot identify a cleverly planned method by the launderers in the future [52, 53].

2.2 AI-based methods

AI is a computer system's ability to simulate human cognitive functions or processes, such as learning and problem-solving. Machine learning is an application of AI [14, 32, 54–56]. It uses mathematical data models to assist a computer in learning without direct instruction [39, 57, 58]. This enables it to learn and improve continuously, based on its own experience or feedback from humans [54, 57, 59]. These days, banks have begun to use AI and machine learning to detect and prevent financial crime [5, 60, 61]. Most financial institutions know that AI is getting faster and cheaper and provides a smarter way to combat financial crime [16, 27]. It is also noteworthy that criminals already use AI [38, 62]. Hence, financial institutions must constantly innovate to stay ahead of criminals by employing multiple techniques that can detect suspicious activity [13, 51, 63]. Within AI and machine learning, two main approaches are supervised and unsupervised [49, 64, 65]. The main difference is that the former uses labelled data to aid in outcome prediction, while the latter does not [49, 65]. In the banking sector, the widely used machine learning types are depicted in Fig. 1.

The widely used supervised algorithms are RF, SVM, and GB. A brief description is discussed below.

Random Forest It is a supervised machine-learning algorithm. It uses randomisation to create a large number of decision trees [66]. Decision trees are the building blocks of the RF model. As its name implies, it is comprised of a large number of decision trees that work together as an ensemble and are typically trained using a bagging method. The bagging method is based on the idea that combining learning models improves the overall result. RF classifier has the advantage of not overfitting the model when there are enough trees in the forest [43, 45]. The greatest strength of the RF is its ability to discover meaningful interactions and non-linear effects of the predictors [46, 66].

Support Vector Machines It is a supervised machine-learning algorithm that can be used for classification and regression. However, it is mostly classification. Because of their distinctive qualities and features, SVM is particularly well suited for binary classification issues like fraud identification. In SVM, each data point is plotted in n-dimensional space (where n represents the number of features).

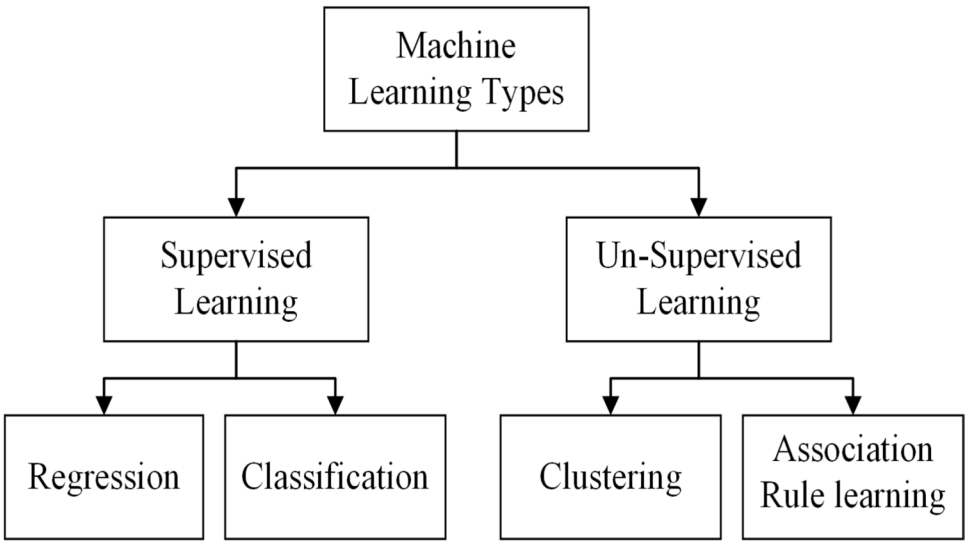


Fig. 1 A high-level taxonomy of the machine learning models widely used in banking-related research

A hyperplane will be identified that distinctly classifies the data points. Since SVMs can extract relevant and significant features from a large number of features, their versatility has made them appealing for detection problems requiring highly unbalanced data sets [13, 50]. Margin optimisation is an additional feature of SVMs that arises from the selection criteria of the hyperplane. It minimises overfitting by optimising the margin of separation between the two classes [13].

Gradient Boosting It is one of the most effective machine-learning techniques. Machine learning errors are broadly classified into two types: bias error and variance error. It is used to minimise the bias error of the algorithm. In GB decision trees, many weak decision trees are combined to produce one strong tree. All the trees are connected in series, and each tree tries to minimise the error of the previous tree.

According to Salehi et al. [67] and Chao et al. [34], unsupervised data techniques help identify patterns and abnormal behaviours that can be instrumental in improving the learning capacity of classification methods. Since machine learning algorithms can adapt as they read new data or patterns, they are widely used and in demand despite the difficulties outlined in Sect. 2.3 [14, 68, 69]. For example, Semmelbeck and Besaw [56] employed an RF classification algorithm to determine the factors that might be significant in determining whether or not a terrorist group is involved in criminal activity. They discovered that a shift in the terrorist group's organisational structure over time is a warning sign for potential money laundering.

According to Badal-Valero et al. [46], Benford's rule should be integrated with machine learning techniques, including logistic regression (LR), decision trees (DT), neural networks (NN), and RF, using the best method available based on whether the data was balanced or imbalanced. They discovered that Benford's law, in conjunction with NN functions, performs best on imbalanced data. At the same time, RF and LR operate best with Benford's law when balancing techniques are used or the data is subjected to the Synthetic Minority Oversampling Technique (SMOTE).

Zhang and Trubey [69] conducted a comparison between the standard Maximum Likelihood Logistic Regression and five supervised machine learning algorithms: DT, RF, SVM, ANN, and Bayes Logistic Regression (BLR). Their findings indicate that ANN outperforms MLLR as a rare event classification algorithm. On the other hand, SVM and RF combined with sampling techniques can also produce similar results. Nevertheless, the difficulty with supervised techniques is that, for the models to produce suboptimal results, the data must be free of biases and errors, the events in the historical data must be correctly characterised, and each input variable must be precisely recognised.

In their paper, Jullum et al. [70] show that XGBoost helps combat suboptimal outcomes because it develops a detection algorithm that predicts the likelihood of money laundering based on the background information of senders and recipients, their past actions, and transaction history. The algorithm treats nonreported alerts, normal alerts, and flagged alerts equally. For supervised algorithms to circumvent this limitation, unsupervised machine learning techniques are suggested as an alternative to boosted algorithms [70]. Data-mining techniques were utilised by Chao et al. to track anomalous activity in trade-based transactions [34].

Another method to surmount the limitations of supervised learning involves merging visuals using the deep learning machine learning algorithm known as the graph learning or clustering algorithm [36]. Li and colleagues suggested utilising FlowScope, a scalable algorithm and a multipartite graph, to visually represent the entire financial transaction flow from the point of origin to the final destination [71]. They claim that FlowScope can identify the fake accounts utilised in the synthetic and real-world datasets more accurately than the state-of-the-art baselines.

Cheng et al. experimented with the group-aware graph neural network-based technique (GAGNN) to identify organised money laundering operations and discovered that this strategy worked well [73, 74]. Detecting organised conduct can benefit significantly from modelling suspicious transactions using group-aware deep graph learning. AML by group-aware deep graph learning.

Afriyie et al. [64] used supervised machine learning to identify the age group and time affected most by credit card fraud using three different categorisation models: decision trees, logistic regression, and RF. The under-sampling technique is used to prevent overfitting and to balance the data. As a result, the RF model outperformed decision trees and logistic regression. Based on these experiments, it was determined that cardholders older than sixty are typically the ones who fall victim to fraudulent transactions, and the time when fraud happens is mostly between 10 pm and 5 am, when most of the banks do not have proper controls for monitoring.

In the process of predicting fraud in the financial markets, Mark Eshwar claims that when RFC (RF classifier) is adjusted using GSCV (Grid search CV), it can produce more accurate findings than logistic regression, DTC (Decision Tree classifier), and the CatBoast algorithms. Mark Eshwar demonstrated that traditional models based on statistical inferences and hypothesis testing are more transparent than machine learning predictions. It needs to be clarified how machine learning models arrive at their predictions. Machine learning models produce prediction scores, yet how these scores are calculated needs to be understood. In contrast to statistical methods, machine

learning focuses on making predictions from performance measures [75]. Eshwar concluded that the model had interpretation bias because no proof of causation could be used against it [67]. Agosto et al. employed an ensemble model based on the principle of combining multiple weak classifiers to build a stronger, high-performing model. For further comparison, they also explored bagging and random forest models, which are part of the same ensemble methodology. The model demonstrated enhanced predictive capabilities and wide applicability. However, this model has been found to have limitations. For instance the evaluation of the model was only limited to available data and hence cannot be generalised. This also leads to an implementation gap in the real world [75, 76].

Paolo et al. conducted a survival analysis aimed at selecting explanatory variables. Their approach utilized mixtures of products of Dirichlet process priors, offering a formal inferential framework to assess the explanatory power of each covariate based on the marginal likelihood associated with the induced partitions of the observations. The proposed model is Bayesian nonparametric, thereby minimizing model specification requirements and enhancing the robustness of the final inferences. The observed strength of this approach is that the model is formulaic in its contribution, which means it provides explicit formulae for marginal likelihoods under various dependent structures, enabling practical application, and the model also supports scalability. But the con side of the model was found to be complex, as the method involved high mathematical and computational complexity, and was not easily interpretable [77, 78].

2.3 Explainable artificial intelligence

One of the primary limitations of AI models is their inability to reason creatively or "think outside the box" [79, 80]. AI can learn over time using pre-fed data and previous experiences, but its approach cannot be creative [81, 82]. An improper algorithm can bring unnecessary errors and wrong results [83, 84]. The truth about AI algorithms is that they rely on data to work effectively [59, 79]. The better the data, the better the outcome [62, 75, 83, 85, 86]. However, poor data quality should not be an excuse to avoid starting the journey to using AI [75, 87]. XAI matters to a range of parties engaged in developing, deploying, and using AI systems [72, 88]. XAI can potentially improve transparency, reduce bias, and increase the dependability and trustworthiness of AI systems by explaining how these systems operate and generate predictions [55, 56, 73, 86, 89]. The target audience of XAI comprises a range of stakeholders engaged in developing, deploying, and utilising AI systems [39, 57, 58, 90]. These consist of stakeholders such as shown in Fig. 2.

- *Scientists, developers, and product owners* AI model construction and training fall under the purview of data scientists and machine learning engineers, who also need to understand how to optimize the models' functionality [72, 83, 91]. They can apply XAI approaches to comprehend the model's behaviour and identify any issues that require fixing.
- *Business decision-makers and leaders* They need to understand AI models' predictions and outputs to act rationally based on the data provided [85, 92].

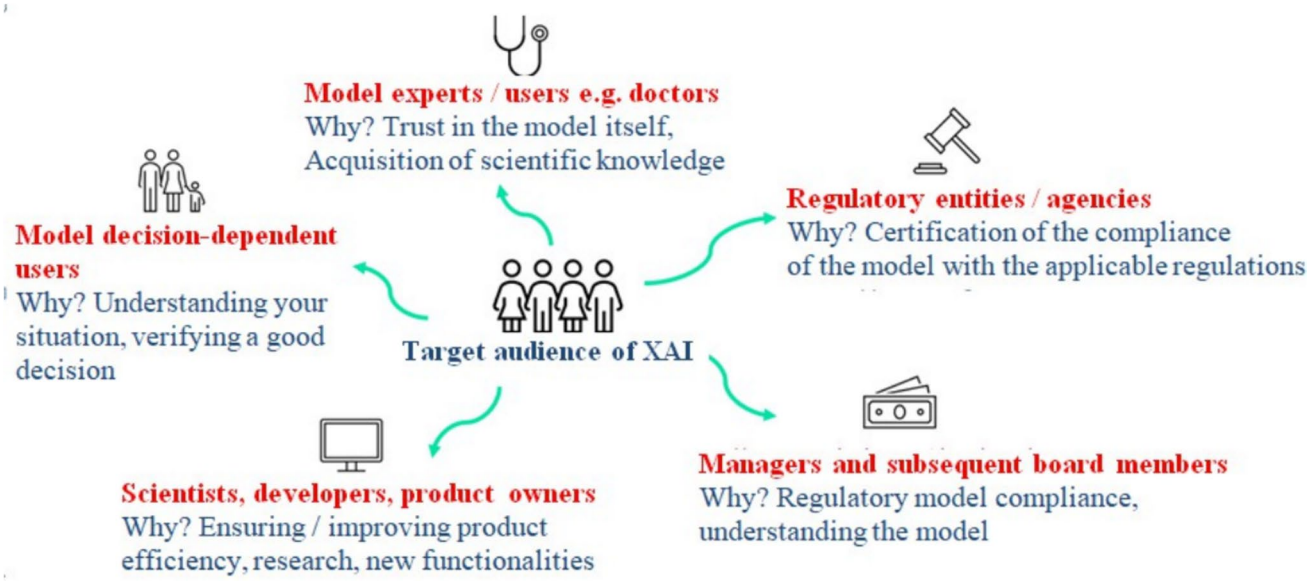


Fig. 2 Target audience of XAI. This figure represents various stakeholders who influence the decision-making. These users need to understand the rationale behind any decision the AI system makes and ensure the decisions are made with integrity and are trustworthy. Figure extracted from [72]

- *Model experts/users* They can use XAI techniques to better understand how the model makes its predictions and to confirm that decisions are based on reliable and accurate information.
- *Regulators and policymakers* It is the responsibility of legislators and regulators to guarantee that AI systems are applied morally and sensibly. They can ensure that AI systems follow all relevant laws and regulations by using XAI techniques to identify any biases or defects in the systems [83, 85, 93].

The primary focus was on research within the domain of financial crime and the corresponding AI methods employed in this field. Table 1 provides the details of the same. The papers were grouped into two categories: Supervised and unsupervised learning. The search was conducted for the period 2016 to 2024. More papers on financial crime were published in 2020. Nearly 60% of researchers used supervised learning, and the remaining 40% used unsupervised learning.

3 Proposed methodology

The proposed methodology consists of three phases: In the first phase, a synthetic dataset is generated, which is required for the research. In the second phase of the paper, supervised machine learning methods, including RF, SVM, and GB, are used to identify suspicious transactions. The questionable transactions, including (1) money laundering, (2) fraud, and (3) a mix of money laundering and fraud alerts, are examined independently during this phase. In the third phase, Shapley Lorenz values, post-hoc XAI approaches, and intrinsic explainer methods are used to better understand the results of the second phase. The possibility of a combined fraud and money laundering risk control framework is explored. Offering a combined framework that might need to be developed to lower AML and fraud rates will benefit Australian regulators. While intrinsic methods provide weight-based explanations rooted in the model's internal mechanics, post-hoc techniques validate these explanations externally. This cross-validation strategy represents a relatively novel approach to explainability auditing. The integration of intrinsic and post-hoc XAI methods offers several key advantages:

Dual-Perspective Insight: This approach delivers a more comprehensive understanding of model behaviour by combining internal model reasoning with external attribution analysis.

Consistency Verification: It facilitates the evaluation of post-hoc explanation fidelity by comparing them against the model's intrinsic decision pathways.

Human-Centric Interpretability: The layered structure of global transparency and local accountability enhances the interpretability of model outputs, making the results more intuitive and actionable for domain experts.

This approach reflects a paradigm shift in XAI: moving from "explaining the output" to "understanding and validating the model's reasoning". It's a step towards auditable, human-aligned AI systems.

3.1 Synthetic data creation

Since the required data is legitimate and sensitive, it must be provided by the organisations. Testing with real data is challenging because of the difficulty associated with sensitivity. Financial institutions store a large volume of private and sensitive information, which is generally highly regulated. The data cannot be shared across different organisations. Various sources have been explored for public, real, and synthetic datasets. However, it was observed that there is no data available that has both fraud and AML data, i.e., the dataset has either fraud or AML but not both. Therefore, a synthetic dataset has been constructed.

The synthetic dataset comprises 50,000 individual bank transaction records, developed using domain-informed rules derived from known AML and fraud typologies. Approximately 12% of transactions were labelled as fraud, 10% as AML, and 6% as both, creating a realistic multi-label distribution. Each record includes over 30 variables designed to simulate typical financial operations while embedding detectable signals for both fraud and AML patterns, such as smurfing, layering, and structuring. Labels were programmatically assigned using if–then rules, reflecting red-flag conditions flagged in FATF and AUSTRAC typologies [108, 109]. The synthetic portion of the data was generated using structured probabilistic modeling to preserve correlations between customer attributes and behavioral patterns for privacy-preserving simulation in finance. Data preprocessing included imputation of missing values, encoding of categorical variables, and min–max normalization of numerical features [110].

Figure 3 provides an overview of the dataset creation process. Code was built to generate a synthetic dataset with fraud and AML features. A Python script was developed to create a synthetic dataset encompassing various financial transaction records. The dataset supports classification tasks for “IsFraud” and “AML_Ind” labels, incorporating diverse and realistic feature values. This dataset includes many features to mimic real-world financial transactions, enhanced using the Faker library for added realism. The structure of the Python script is as outlined below:

Table 1 Various machine learning models used in financial crime

S. No:	Category	Problem	Method	Results/Any other issues	Data	Year	Ref#
1	Supervised Learning	Detection of probability of money laundering transactions	Random Forest, Naïve Bayes classifiers	The NB classifier correctly predicted money laundering transactions 39.4% of the time, while the RF classifier predicted 50.3% of the time, which implies that RF outperformed the NB classifier	Real	2024	[11]
2		Identification of fraudulent identities	SVM, KNN, Naïve Bayes, Tensorflow model	• The Tensorflow model demonstrated how it offers more functionalities (adjusting and setting new parameters for optimization, bias weight, etc.) and parameters that can altered to improve the model than just the traditional method of training and testing of the datasets in other ML models • The TensorFlow model resulted in the best results when the 80–20 sampling technique was used compared to SVM, KNN, etc	Public dataset	2023	[32]
3		Detection of organized money laundering transactions	SVM, GB, Graph neural network-based approach (GAGNN)	The group-aware graph neural network-based approach (named GAGNN) is best for grouped (gang) behaviour in detecting money laundering criminals	Real	2023	[73]
4		Detection of credit card fraudulent transactions	Decision trees, logistic regression, RF	• RF was identified as the most suitable model with an accuracy of 90% • Undersampling -The balancing technique ensured that the model did not favour the major class solely • Cardholders over 60 years old are identified as the majority of victims	Synthetic	2023	[64]
5		Detection of fraud transactions	Logistic regression, Decision Tree classifier, Random Forest classifier, Gradient boosting (CatBoast)	Below are the challenges that were observed: • The ML prediction models used were less transparent than hypothesis testing and statistical inferences • The process by which ML models reach predictions was unclear • However, ML techniques offer opportunities for prevention	Real	2022	[68]
6		Detection of fraud transactions	Random Forest, Decision Tree, GB, KNN (Nearest Neighbour)	• For commercial entities, in the multi-class classification, RF outperformed the other classifiers with 93% accuracies, followed by DT with 90% accuracy	Real	2022	[26]
7		Detect suspicious transactions and reduce false positives in AML	Random Forrest, Amaretto (active learning system for anomaly detection)	• Amaretto's model provided good performance within a short time frame and with minimal input from SMEs	Synthetic	2022	[94]
8		Watch-list filtering automation to prevent false positives	SVM, Naïve Bayes, Decision Trees	• SVM outperformed the other algorithms in the process of watch list filtering	Synthetic	2021	[95]
9		Predict illicit transactions in the Bitcoin transaction graph	Graph convolutional network intertwined with linear layers	• The concatenation of features derived from GCN, and the latent representation of a linear layer boosted the model's performance	Elliptic	2020	[96]
10		Detecting deemed illicit behaviour in AML with high confidence	Logistic Regression, Extreme Gradient boosting trees, Catboost methods	• Two-layered model to address Scalability (first layer using LR) and imbalance resistance (second layer with GB	Real	2020	[33]
11		Detect illicit accounts involved in money laundering over the Ethernet blockchain	Extreme Gradient Boosting	• Illicit behaviour over the Ethereum network at an account level was detected efficiently and effectively using the XGB model, which takes input from past transactions and account properties	Public Dataset	2020	[97]

Table 1 (continued)

S. No:	Category	Problem	Method	Results/Any other issues	Data	Year	Ref#
12		Detect money laundering transactions using ML	Logistic regression, decision tree, random forest, XG Boost, Auto Encoder, DNN	• Technical and empirical evidence of the AML framework with new methodologies such as AI and network science were developed • Identified the necessity to fill a gap between ML dynamics and current tools of AML because of these tools as a result of a static analysis after several days or months of making the financial transactions	Real	2020	[82]
13		Detect suspicious money laundering transactions	XG Boost algorithm	• The XGBoost framework was efficient in predicting the probability that a transaction should be reported	Real	2020	[70]
14		Detect suspicious money laundering transactions	Naïve Bayes Classifier	• The highest accuracy (81.2%) was achieved using the Naïve Bayes classifier in terms of detection as legal or illegal	Synthetic	2020	[98]
15		Identification of money laundering in the Bitcoin blockchain in the presence of label scarcity	LR, RF, XGB	• Unsupervised anomaly detection methods proved to have poor performance, and anomalies in the feature space were not indicative of illicit behaviour • It was demonstrated that experiments conducted on synthetic data can be misleading, and the importance of conducting experiments on real data was emphasized	Elliptic	2020	[99]
16		Identification of suspicious transactions and categorizing the type of finance crime	SVM, Linear Regression, Multilayer Perceptron	• MLP outperformed with an F1-score of 90.48% compared to SVM and LR, with F1- scores of 87.89% and 87.59%, respectively	Synthetic	2019	[100]
17		Suspicious money laundering transaction detection	Bayes Logistic Regression, Decision Tree, Random Forest, SVM, and ANN	• ANN model was observed to be a better model performing compared to the others when under-sampling and over-sampling techniques were used	Real	2019	[69]
18		Suspicious money laundering transaction detection	RF, MinMaxScaler Method	• Accuracy was achieved at 90% for the RF classifier when min mix scalar transformation was used	Synthetic	2018	[101]
19		Detect money laundering groups	SVM and RF	• RF performed slightly better than SVM. The average recall of both SVM and RF was quite low. However, both models exhibited an extremely high precision	Real	2017	[49]
20		Improve suspicious transaction signalling process by client profiling	PART algorithm that implements decision tree, best leaf technique	• Rules generated by algorithms PART and J48 were successfully able to identify the profiles of standard customers and customers with varying risk categories	Real	2016	[87]
21	Unsupervised Learning	Detect suspicious transactions and reduce false positives	Knowledge management systems	• The importance of various knowledge management systems was emphasized • The importance of systems integration to combat was highlighted	Synthetic	2022	[7]
22		Identify suspicious behaviour patterns for risk profiling	Client risk profiling, Behaviour flagging	• Risk rating practice as part of KYC onboarding customers was studied using K means clustering	elliptic	2023	[52]
23		Compare the risk management of cryptocurrency	Clustering, recursive bisection, Reinforcement Learning	• Reinforcement Learning yielded a high performance compared to other techniques	Real	2022	[102]
24		Suspicious money laundering transaction detection	Core Decision Tree and Clustering Algorithm	• The inclusion of additional non-transactional variables and filtering by the indicators of abnormality lead to a lower number of false positives	Real	2021	[103]
25		Identify fraudulent financial transactions	Outlier detection methods, visual analytics method	• Different sets of dashboard views (visualization techniques) with abstraction levels were used to identify outliers • The SVM One classification, HBOS, and XBOS outperformed in outlier detection	Real	2020	[104]

Table 1 (continued)

S. No:	Category	Problem	Method	Results/Any other issues	Data	Year	Ref#
26		Identify anomalies in a set of transactions of a non-banking correspondent	Isolation forest, One-class SVM	• Isolation Forest performed better compared to SVM	Real	2020	[60]
27		Detect hidden networks of money launderers	Isolation forest, One-class SVM	• Isolation forest performed better than the One class SVM	Synthetic	2020	[65]
28		Identify suspicious transactions in cryptocurrency	Expectancy Maximization, RF	• RF resulted in high precision in the detection of abnormal transactions	Elliptic	2019	[105]
29		Detect anomalies considering the regular transaction patterns Auto Encoder and PCA were	examined separately	Auto Encoder algorithm detected more anomalies with high accuracy and had a power nonlinear generalization • The limitation identified was the evaluation of results against the business objectives	Real	2017	[106]
30		Identify suspicious money laundering transactions	Isolation forest, One-class SVM, EM. Gaussian Mixture Models	• OCVSVM had a degraded accuracy when too many anomalies were present compared to the other methods	Public	2017	[35]
31		Detection of money launderers' gangs	Community detection using a temporal-directed Louvain algorithm	• The TD Louvain algorithm not only made the distribution of fragmented nodes more structured but also helped to divide a large complex network into smaller groups with more explicit meanings	Real	2017	[81]

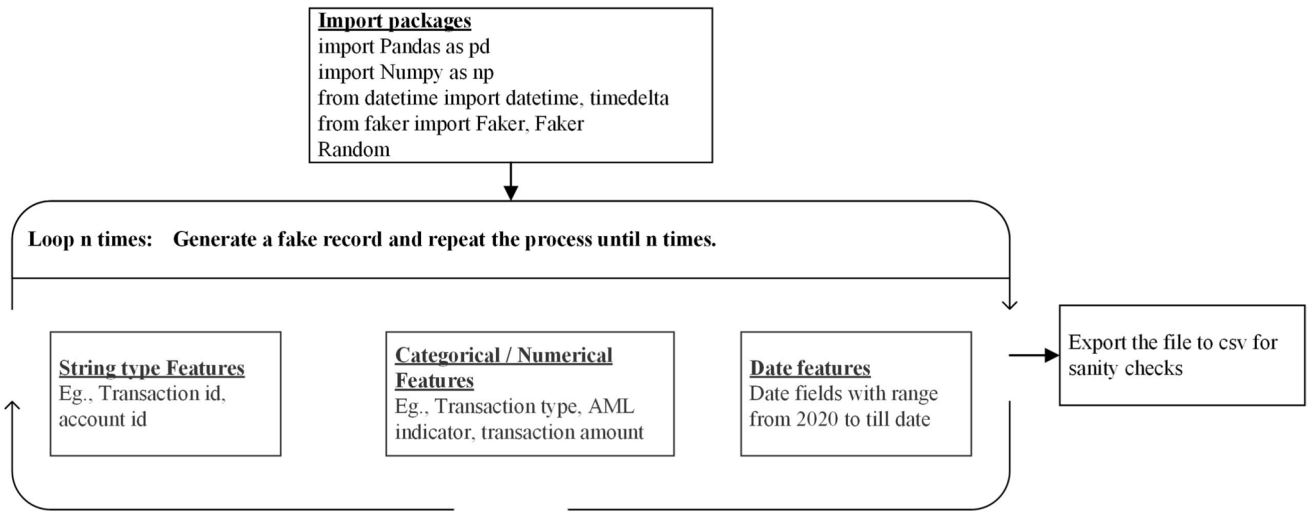


Fig. 3 Synthetic data generation workflow

- *Importing Libraries* Pandas, NumPy, and Faker have been imported.
- *Defining Utility Functions* Functions for generating dates, IDs, and diverse values have been defined.
- *Generating Synthetic Data* Loops and conditional logic have been used to fill each record with appropriate feature values.
- *Saving the Dataset*: The generated dataset has been outputted to a CSV file.

The required variables are broadly classified as Transactional, Account, Customer, Behaviour pattern and risk indicators, Compliance and historical data, Technological Footprints, and Fraud Typology. The generated file was carefully examined to confirm the presence of all required variables. The process started with making rules for creating data. These rules included how to make accounts and customer profiles, as well as how to make transaction amounts, types, and frequencies. The goal was to make a system that could simulate real financial transactions by considering

factors such as the amount of risk and the way transactions usually happen. After that, Python scripts for creating data were implemented using Pandas, NumPy, and Faker libraries. These scripts were the practical way to turn the rules that had been defined into code that could be used. Now that the scripts were in place, transaction data could be made. Random number generators were used in this phase to figure out the amounts and frequencies of transactions. It was important to ensure that the dataset included transactions that did not fit the norm. These transactions could be signs of possible money laundering. The simulation of transaction dates and times also tried to show how time works in real life.

Account and customer data were added to the dataset at the same time. This meant making account information and ensuring there were a variety of account types and locations. It was also possible to make customer profiles, which put customers at risk levels and describe how they usually do business with each profile. Anomalies and patterns were added to the dataset to make it seem more real. Anomalies included sudden increases in the amounts or frequency of transactions, while patterns looked like the different steps of laundering money, such as layering and integration.

Key parts of the script include the functions to generate transaction IDs using Faker and methods to create dates within a realistic range. It was very important to check and improve the data generated to make sure it was accurate and consistent. During this step, the dataset was carefully looked over, and any changes that were needed were made to the generation rules to make the dataset more in line with real-life cases of money laundering. Table 2 lists the Python libraries that were used. The Python script generated a synthetic dataset that encompassed various financial transaction records. The dataset supports classification tasks for “IsFraud" and “AML_Ind" labels, incorporating diverse and realistic feature values. The generated dataset has been reviewed for consistency and realistic variability. Parameters controlling the diversity and distribution of feature values have been adjusted, if necessary, to better align with expected real-world data characteristics.

Table 2 Libraries required for the study

Library	Description of the library		
Faker	Generates Fake Data XAI-Alibi	XAI Library	
Pandas	Data analysis library for importing Excel file NumPy	Used for mathematical and statistical functions Seaborn	Data visualisation
Matplotlib	Graphical plotting Sklearn	Machine Learning library	
SHAP	XAI—Shapley explainer library LIME	XAI LIME explainer library	
Dash	Analytical web application library		
LightGBM	Classification and regression library		

The final output, a CSV file named as synthetic_fraud_aml_dataset.csv containing the synthetic dataset, is ready for use in developing and testing machine learning models for fraud and AML detection tasks. The dataset closely mimics real-world financial transactions and is a valuable resource for training and evaluating predictive models. The author considered predictive utility and statistical similarity dimensions for the evaluation of the synthetic data credibility. As part of statistical similarity, the paper has referenced various public and synthetic datasets available to factor in the features that a fraudulent and money laundering transaction will have [7, 32, 81, 94, 95, 97, 98, 100, 101]. For the sake of predictive utility, the F1-score was calculated to check and assess how close the dataset is to real data. RF, SVM and GB algorithms were used, and the f1 score ranges from 0.54 to 0.78, and this has been compared to F1-scores calculated in the real world. As part of mimicking real financial transactions, particularly about evolving criminal tactics, the data has been replicated by layering money laundering (smurfing, structuring, etc.), credit cards, money mules, geo-locations, high-value structuring not consistent with previous behaviour, high-risk countries, structuring to evade detection threshold, etc.

Using machine learning models like RF, SVM, and GB for financial crime detection, particularly fraud and AML, is widely supported in academic and industry literature. An empirical comparison, summary and justification for their use in banking are mentioned below:

Random Forest (RF)

Strengths:

- Handles imbalanced datasets well, which is common in fraud/AML.
- Robust to noisy or missing data.
- Provides feature importance, aiding interpretability.

Empirical Use:

- RF was a top performer in baseline models [111].
- RF is used as one of the most used algorithms in fraud detection systems due to its accuracy and robustness [112, 113].*Support Vector Machines (SVM)*

Strengths:

- Excellent in high-dimensional spaces.
- Performs well with limited samples and complex boundaries.
- Effective when fraud patterns are nonlinear and sparse.

Empirical Use:

- Used to detect fraudulent transactions with high precision, showing its effectiveness on small datasets [70].

Table 3 Summary of comparison of various models performance

Criterion	Random forest	SVM	Gradient boosting
Accuracy (fraud)	High	High (small data)	Very high
Scalability	Medium	Low	High
Imbalanced data handling	Good	Fair (with tuning)	Excellent (with tuning)
Interpretability	Moderate	Low	Moderate (e.g., SHAP)
Industry use	High	Moderate	Very high

Table 4 Required key attributes

Feature	Description
Irregular_Behaviour_Flag	Indicates whether a transaction deviates from the customer's usual pattern, suggesting potential fraud
Anomaly_Flag	Marks transactions as anomalies based on outlier detection algorithms, signifying transactions that are unusual and potentially fraudulent
AML_Ind	While primarily for AML, this indicator can also be relevant for fraud detection, as money laundering activities may overlap with fraudulent transactions
Compliance_Flag	Reflects whether the transaction complies with regulatory standards, where non-compliance might be associated with fraudulent intentions
Previous_Fraud_Reports	Indicates whether there have been previous fraud reports associated with the customer or account, suggesting a higher risk of recurring fraudulent activity

- SVM, when compared with neural networks and decision trees was found to have better generalisation in AML pattern detection [43].*Gradient Boosting (GB/GBM/XGBoost/LightGBM)*

Strengths:

- Highly accurate with strong generalisation.
- Flexible: Handles missing data and custom loss functions.

Empirical Use:

- GBM was found to outperform traditional methods on extremely fraud data [114].
- GB models outperformed all others in terms of AUC and F1-score in large-scale fraud detection datasets [114].

Summary is listed in Table 3.

When it comes to detecting fraud, Boolean features are essential because they provide binary information about various aspects of transactions, accounts, and user behaviour. Some of these features include Irregular_Behaviour_Flag, Anomaly_Flag, Compliance_Flag, and Previous_Fraud _ Reports. The Irregular_Behaviour_Flag highlights transactions that deviate from a customer's typical pattern and may indicate fraud. The Anomaly_Flag is set for transactions that, after a thorough investigation, are discovered to differ significantly from accepted norms, implying possible fraud. The Compliance_Flag determines whether transactions comply with financial transaction rules. Non-compliance is frequently associated with increased fraud risk. Finally, Previous_Fraud _Reports displays whether a customer or account has a history of associated fraud, indicating a higher risk of future fraudulent activity. The key attributes are summarised in Table 4. The generated synthetic dataset is ready for further training and experimentation on the XAI models in the subsequent sections.

3.2 Traditional machine-learning models

The synthetic data generated, as explained in the above section, is in the CSV file format. Each row contained the fraud identifier, AML identifier, and the 'AML and fraud' identifier. Three supervised learning algorithms— RF, SVM, and GB (via LightGBM)—were employed due to their strong track record in financial anomaly detection. RF was configured with 100 trees and a maximum depth of 10 to balance complexity and interpretability. SVM used an RBF kernel with $C = 1.0$ to handle non-linear boundaries. GB was trained with a learning rate 0.05 and early stopping set to 20 rounds.

Prior to training, numerical features were standardized, and categorical variables were one-hot encoded. Class imbalance was handled using class weighting in SVM and scale_pos_weight in GB. A fourfold stratified cross-validation setup was adopted to ensure stable evaluation across minority classes (fraud/AML). Model evaluation focused on F1-score, precision, recall, and AUC due to their relevance in imbalanced classification settings. The high-level process of how the data is being selected for experiments is shown in Fig. 4.

The sequences of steps are explained below:

- The synthetic dataset generated will be loaded into a Python data frame.
- The categorical and numerical features are defined.
- Features that are not required for the model are removed.
- The pre-processing pipeline is defined with a standard scaler for numerical values and one hot encoder for categorical values.
- Target features are defined. This depends on the experiment conducted. For example, AML_Ind will be the

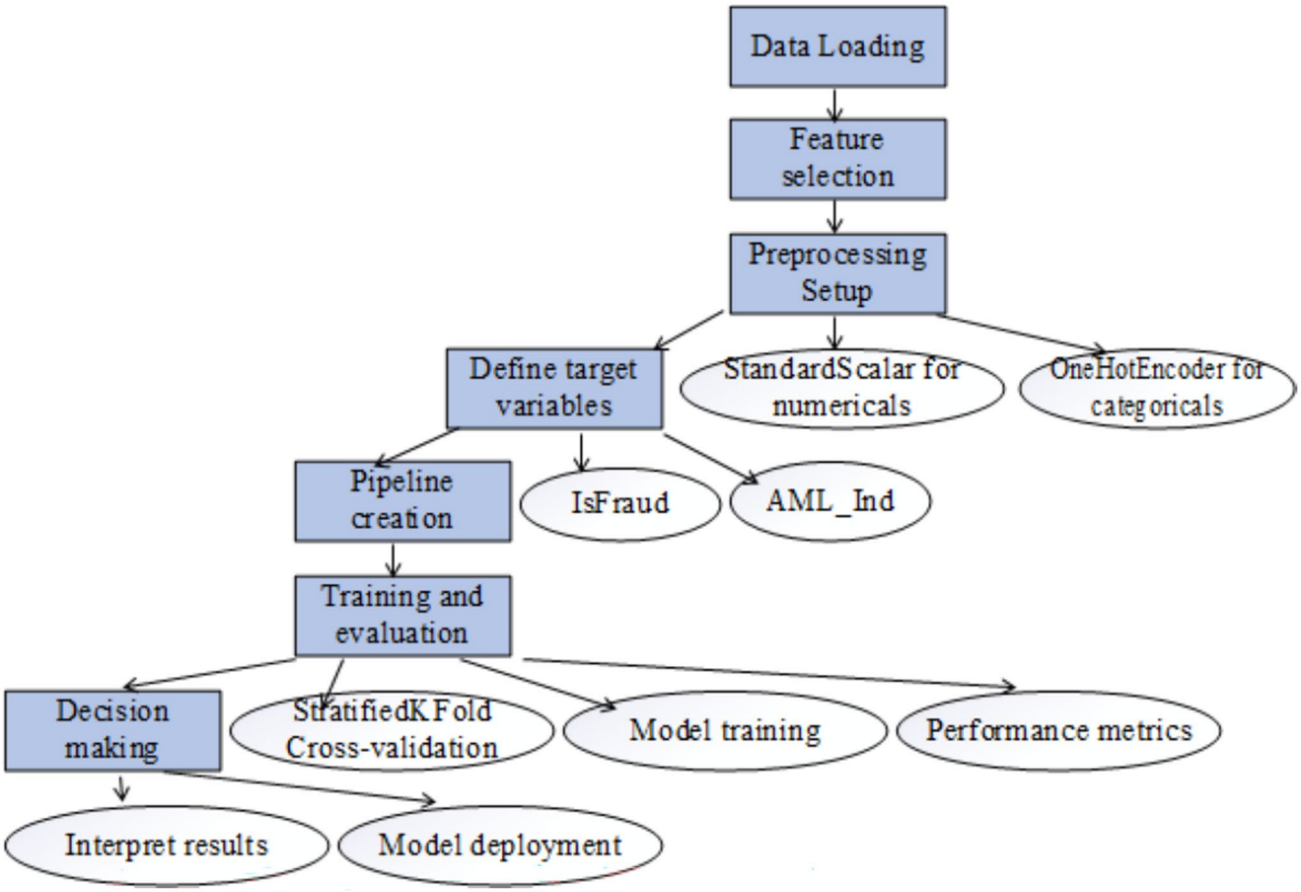


Fig. 4 The proposed pipeline for model training

Table 5 Target variable identification. The target variable is identified depending on the type of suspicious record. The target variable for building the AML transactions model is AML_ind. Similarly, the target variable for building the fraud transactions model is'Isfraud.'

Detection type	*Target variable*
AML transactions	AML_Ind
Fraud transactions	IsFraud
Both AML and fraud transactions	AML_Ind and IsFraud

target variable for detecting AML transactions. Similarly, for the detection of fraudulent transactions, 'IsFraud' becomes the target variable.

- The model is then trained and evaluated using the stratified fourfold cross-validation scheme.
- The F1-score is calculated to gauge the performance of the model.
- ROC-AUC curves were plotted for fraud and AML transactions with RF, SVM and GB classifiers to further assess the relative closeness to real data.

Nine experiments were conducted to analyse the data, obtain the performance of each model, and compare the performance of each model. The sequence of steps is repeated for all nine experiments. Target variables for the model are identified as per the experiments, as shown in Table 5. The F1-score was calculated for all nine experiments conducted and was recorded in Table 6.

Table 6 F1-score comparison. The table represents the statistics on F1-score for the models built using traditional machine learning algorithms such as RF, SVM and GB

ML algorithm	AML (F1-score)	Fraud (F1-score)	Combined AML and fraud (F1-score)	*p* value
RF	0.63	0.66	0.62	0.014 (significant)
SVM	0.56	0.71	0.65	0.062 (marginal)
Gradient boosting	0.54	0.78	0.66	0.009 (significant)

To contextualize our results, we benchmarked the performance of our selected models (RF, SVM, and GB) against traditional classifiers such as logistic regression, naïve Bayes, and decision trees, as reported in prior works (see Table 1). For example, Afriyie et al. [64] and Badal-Valero et al. [46] demonstrate that logistic regression and decision trees generally achieve lower F1-scores (ranging from 0.39 to 0.62) in fraud and AML detection tasks. In comparison,

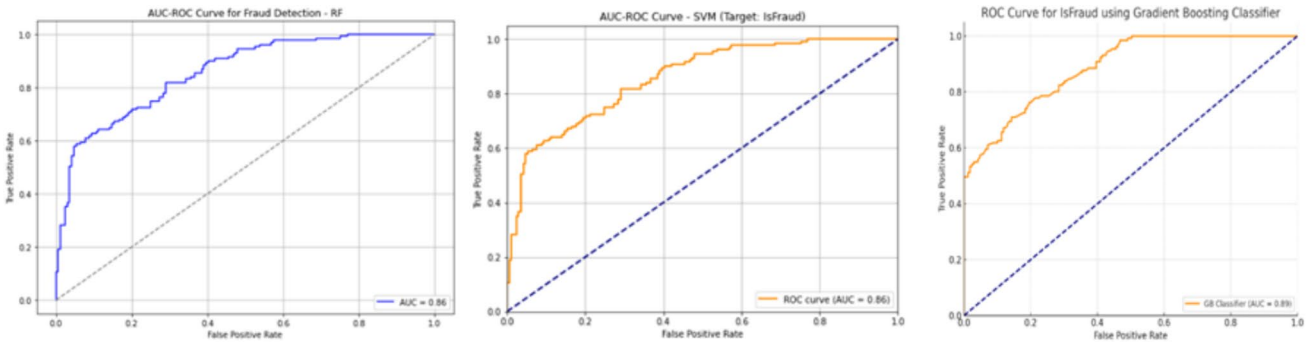


Fig. 5 AUC-ROC plot for fraud transactions with RF, SVM and GB classifiers

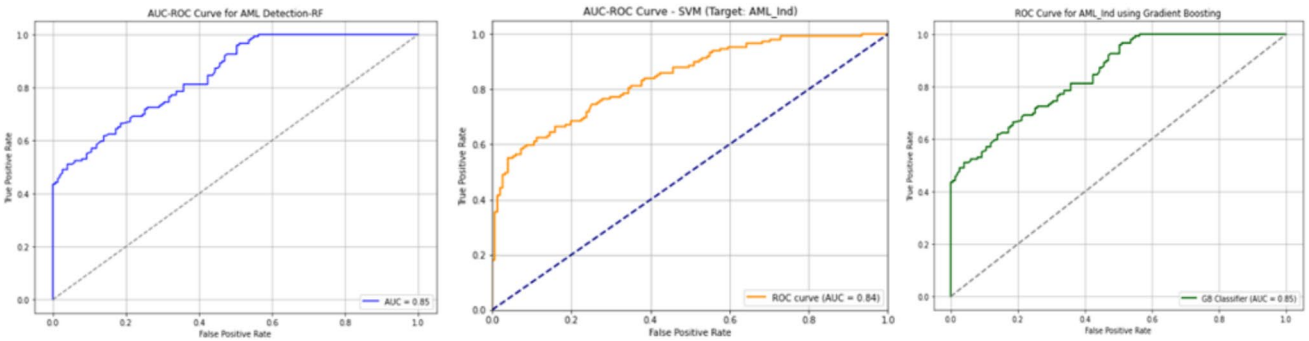


Fig. 6 AUC-ROC plot for AML transactions with RF, SVM and GB classifiers

our models achieve F1-scores between 0.54 and 0.78 (Table 6), indicating consistent improvements over baseline methods. This aligns with recent studies in the domain [68, 69, 94], where ensemble and margin-based methods (RF, SVM, GB) outperform conventional classifiers. Thus, in addition to achieving competitive predictive performance, our approach incorporates explainability, which is sometimes overlooked in conventional models.

ROC-AUC curves were plotted for fraud and AML transactions with RF, SVM and GB classifiers to further assess the relative closeness to real data [115–117]. From the AUC-ROC plot shown in Figs. 5 and 6, the AUC factor ranges from 0.84 to 0.85 for AML transactions, and from 0.86 to 0.89 for fraud transactions. AUC value of 0.80 is considered an excellent performance metric.

Fraud

AML

Across the fourfold cross-validation results, paired t-tests were performed to assess the robustness of the reported enhancements. RF produced statistically significant improvements compared to logistic regression and decision trees ($t=3.12$, $p=0.014$). Likewise, GB performed significantly better than these baselines ($t=3.45$, $p=0.009$). Significant gains were also demonstrated using SVM, with findings close to significance ($t=2.08$, $p=0.062$). These outcomes confirm that the performance gains of our ensemble and margin-based models are unlikely to be due to random variation. Notably, the robustness of results across folds further strengthens confidence in our framework's generalizability.

3.3 XAI methods

This paper explored the following using the proposed synthetic dataset, as explained in Sect. 3.1.

- Post-hoc explanation:

Feature importance This method is model-agnostic and not restricted to any specific models. The advantage of this model type is that the interpretability technique is not limited to particular models. LIME and SHAP techniques are used in this paper A feature optimization technique was employed during the experiments as part of the intrinsic explanation approach. Shapley-Lorenz values, developed by Giudici and Raffinetti, provide a means to quantify the contribution of individual explanatory variables to the predictive accuracy. They are significant because, in contrast to ordinary Shapley values, which can be challenging to compare across various models or response variables, they offer a normalized, interpretable measure of variable importance. We selected SHAP and LIME as post-hoc explainers because of their wide adoption, model-agnostic nature, and ability to provide both global and local interpretations [85, 118].

Because standard Shapley values are not normalized, they are reliant on the response variable's unit of measurement. For this reason, it won't be easy to compare the importance of variables across different models or datasets. Shapley-Lorenz values are simpler to understand and compare across many models and response variables because they employ Lorenz zonoids, which are standardized on a scale of 0 to 1. Unlike standard Shapley values—which are unbounded and unit-dependent—Shapley-Lorenz values are normalized between 0 and 1 using Lorenz zonoids [110]. This makes them particularly suitable for comparing variable influence across different models, datasets, or outcomes [110, 119]. In this study, we apply them to evaluate and rank the contribution of features in fraud and AML prediction tasks.

- A model is intrinsically explainable if it is 'by-design' interpretable. Regression and decision trees are common examples whose computational structure makes decision-making transparent [84]: Decision tree and logistic regression.

A summary of experiments that will be conducted as part of XAI is shown below in Table 7.

RF is chosen as the classifier as it is widely used across many financial institutions because of its simplicity and ease of implementation.

Feature importance with Feature optimisation Experiments are also done by including feature optimisation techniques in the feature importance methods. This checks if the performance varies when feature optimisation techniques are used. The high-level process of how the data is being selected for experiments is shown in Fig. 7.

The sequences of steps are explained below:

- Required libraries are installed and imported.
- The synthetic dataset generated will be loaded into the Python data frame.

- As part of data pre-processing, unwanted columns are dropped, and categorical values are encoded.The target or label is defined.

Table 7 Summary of XAI experiments. For each financial crime type, both post hoc and intrinsic explainers are used to study the interpretability of the model

Financial crime type	Post-hoc	Intrinsic		
AML	LIME	SHAP	Decision tree	Logistic regression
Fraud	LIME	SHAP	Decision tree	Logistic regression
AML and fraud	LIME	SHAP	Decision tree	Logistic regression

- RFE with LightGBM model is used for feature selection. Setting the model to select the top 10 contributing features.
- The percentage of test and train split is set.
- The model is trained using the Light GBM classifier.
- The model is evaluated for its performance.
- Post hoc explanations are applied using LIME and SHAP techniques.
- The results are interpreted.
  The two methods below were used for the feature selection process.
- *Wrapper method to achieve high performance* Recursive feature optimization with a decision tree model was used to select required features, and out of the contributing features, the top 10 were chosen.
- *Domain-driven method for high interpretability* The Author has used his experience in the AML/fraud domain to identify red flags and risk indicators. Some of the features are manually selected based on typologies.

Intrinsic Explanation An experiment was also conducted for Interpretable XAI by choosing the top 10 contributing features. This is done as a pilot that can be used for situations like major crime investigations, where the data is small compared to normal money laundering and fraud cases. However, the technique was applied to interpret fraud and money laundering cases in this case. The high-level process of how the data is selected for experiments is shown in Fig. 8.

The sequences of steps are explained below:

- Required libraries are installed and imported.
- The synthetic dataset generated will be loaded into the Python data frame.
- As part of data pre-processing, unwanted columns are dropped, and categorical values are encoded.
- The target or label is defined.
- The percentage of test and train split is set.
- The model is trained using a Decision tree and Logistic regression.
- Both the models are compared for performance.

The model with the highest performance is further interpreted using summary graphs.

4 Experimental results and discussions

4.1 Fraud detection

A set of experiments are conducted to study the performance and understand the interpretability of the model for fraud

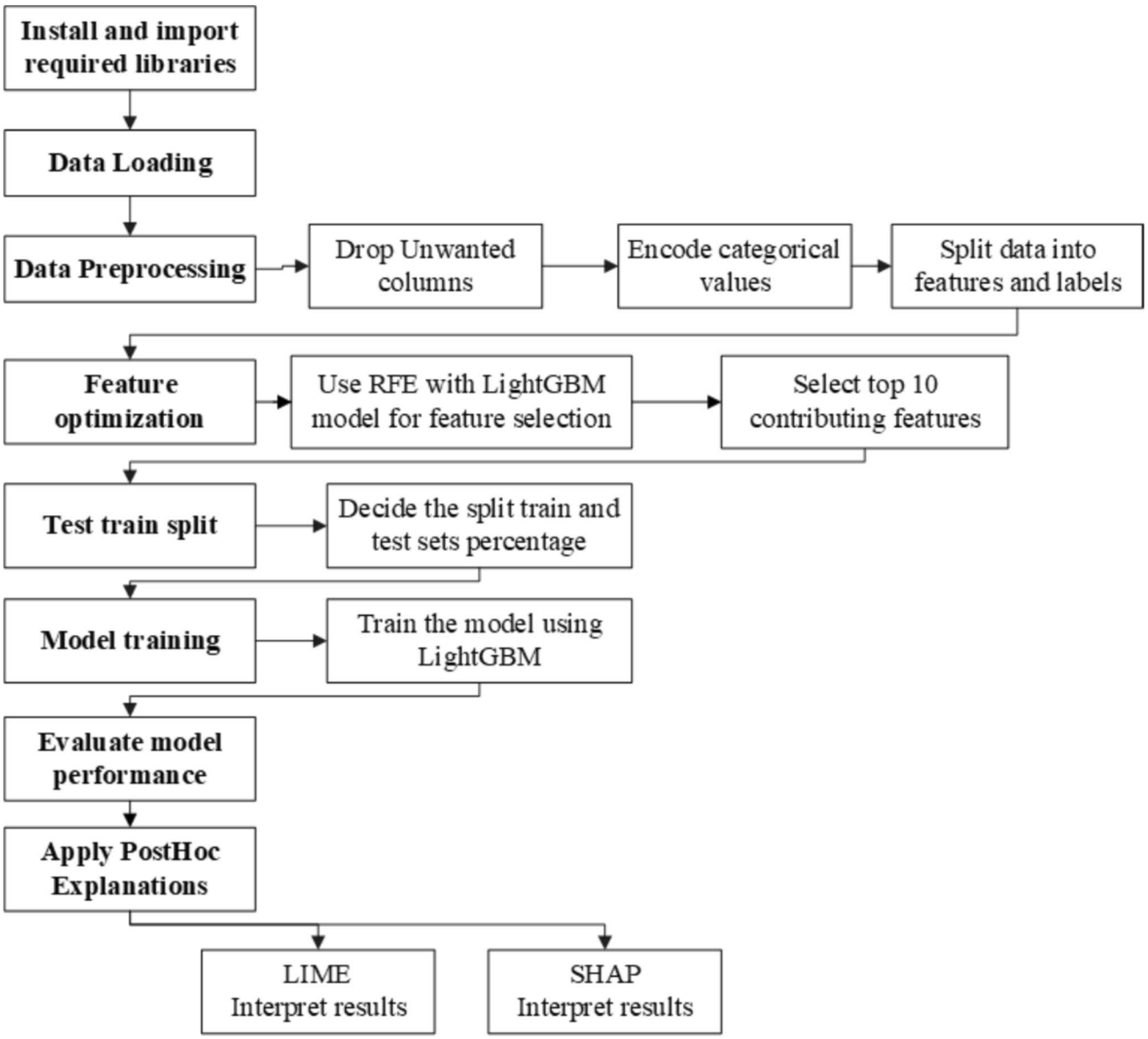


Fig. 7 High-level process flow of post hoc method. Feature optimization using RFE with LightGBM classifier is used with parameters 'random state' as 42, and 'features to select' as 10 was set. The optimized data was split into training and test sets with 0.8 and 0.2, respectively

transactions. Feature importance and intrinsic explanation methods were used.

SHAP and LIME were used in the experiment. The feature optimization method was applied to minimize the number of features, reduce computational complexity, and enhance the efficiency of the machine learning model. The recursive feature elimination method was applied along with lightGBM to limit the number of features required for the training dataset. RFE aided in limiting the number of irrelevant or redundant features and avoided the building of overly complex interpretable models. A lightGBM classifier is used to make the model training faster. The model was set to consider the top 10 contributing features for interpretability. The model is then trained on the lightGBM classifier for optimised features. The accuracy of the model was evaluated and found to be 49%.

For the SHAP summary plot, the x-axis represents the Shapley value, the y-axis represents the value of the feature, and the colour represents the value of the feature from low to high. The top 10 contributing features are plotted in the SHAP graphs in Fig. 9. The feature 'transaction amount' appears to be the top contributing feature in deciding whether the transaction made is a fraud or not. The next significant contributing feature is the 'risk score'. The third significantly contributing feature is the 'Transaction anomaly

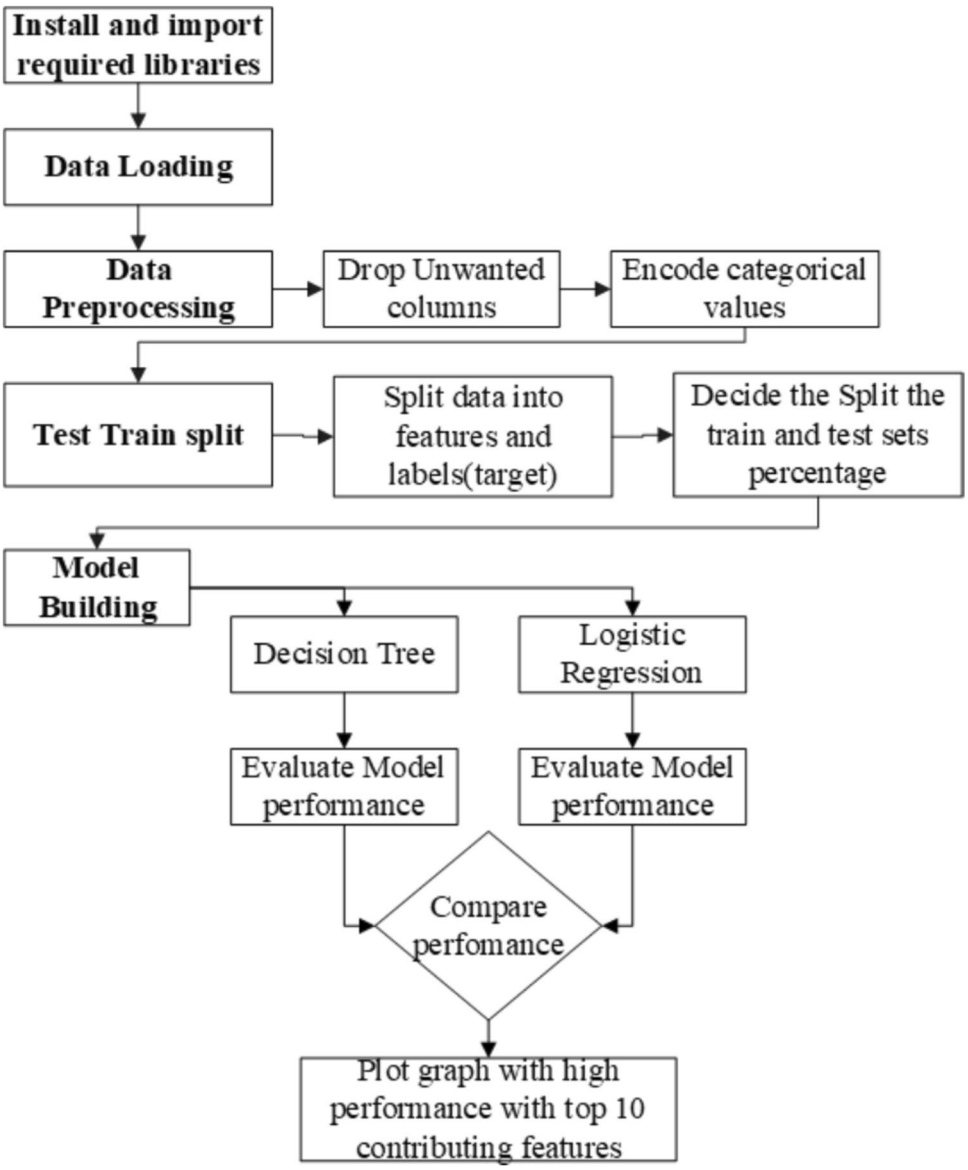


Fig. 8 High-level process flow of Intrinsic explanation. The flow diagram explains that feature optimization is done using RFE. With optimized features, the models were separately built using a Decision tree and logistic regression. The accuracy of both models is evaluated and compared

score', which represents the outliers in the behaviour or transactions made by the customer.

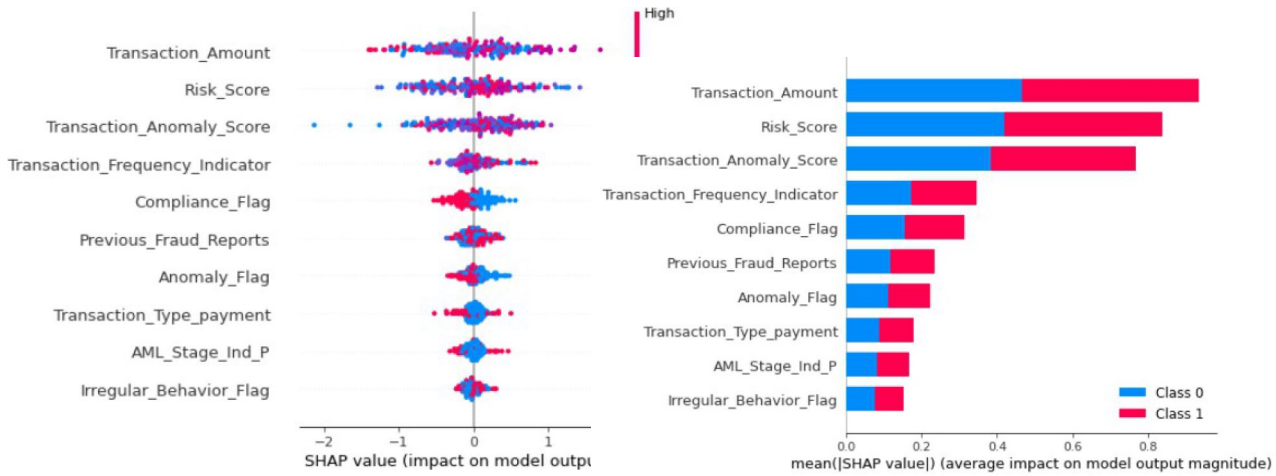
For the LIME explainer, it appeared from Figs. 10 and 11 that the top three contributing features to the model are the 'Compliance flag', 'Anomaly Flag', and 'Transaction frequency indicator'. LIME explainer interpreted the transactions to be an AML record that has:

- deviated from the compliance policy
- outlier or an anomaly transaction
- multiple transactions made not in line with the normal frequency.

Shapley-Lorenz values Table 8 summarizes the findings of the calculation of Shapley-Lorenz values.

The top four features—fraud_AML_Ind, isfraud, risk_score, and transaction_anomaly_score—account for 85% of the overall impact. This tells us the feature contribution is unequal. The top two indicators are related to fraud, the next two major contributing features are the risk score and transaction anomaly score. The computed Ginni coefficient value is approximately 0.58, indicating moderate-to-high inequality. This indicates the model relies heavily on a few features to make predictions, which is typical in domains like fraud detection, where certain flags or risk scores dominate. The inference from the Table 8 is that banks should focus on practical ways of risk rating customers and improving transaction anomaly scoring engines to reduce the impact of fraudsters.

Intrinsic Explanation Decision tree and logistic regression classifiers were used to interpret the model using



(a) SHAP value for model output (b) Mean SHAP value

Fig. 9 SHAP plots for fraud. The figure shows the SHAP plots for the top 10 contributing features of the model. The features 'transaction amount' and 'Irregular behavior flag' are the model's top and least contributing features, respectively

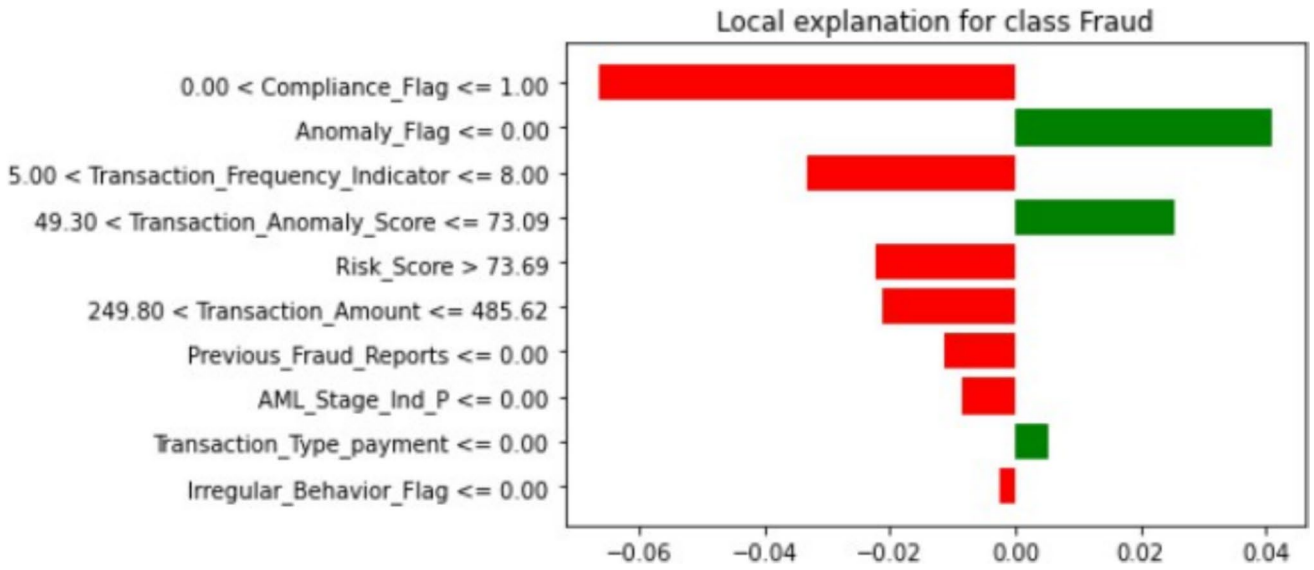


Fig. 10 LIME: Feature importance for fraud. The figure shows the LIME plot for the top 10 contributing features of the model built. The features 'Compliance flag' and the 'Irregular behavior flag' are the top and the least contributing features of the model, respectively

Intrinsic explanation. Features were first optimised using RFE and a decision tree to make it faster and to consider significantly contributing features to the model. The model was set to consider the top 10 contributing features for interpretability. With the optimized features, the model was trained using the decision tree. The accuracy was found to be 70%. With the same optimized features, the model was once again trained using logistic regression, and the accuracy was found to be 70%. In this case, the accuracy was the same when decision tree or logistic classification classifiers were used, as seen in Fig. 12.

A coefficient plot was generated for the logistic regression classifier. From the plot, as seen in Fig. 13, it was studied that the 'AML indicator' and the 'Both Fraud and AML' indicators are strongly correlated, while the other features are weakly correlated.

Comparison

A comparison of the outcome of feature importance has been studied for fraud models as seen in the Table 9. A consistent presence of features such as 'Transaction amount', 'Risk score', and 'Transaction frequency indicator' suggests their fundamental importance in detecting fraudulent transactions.

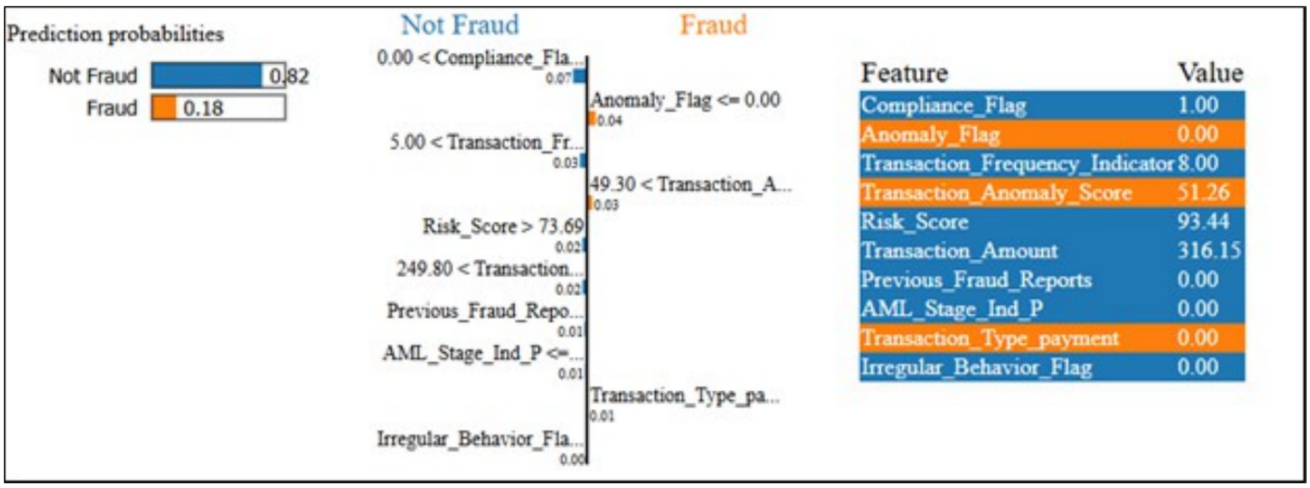


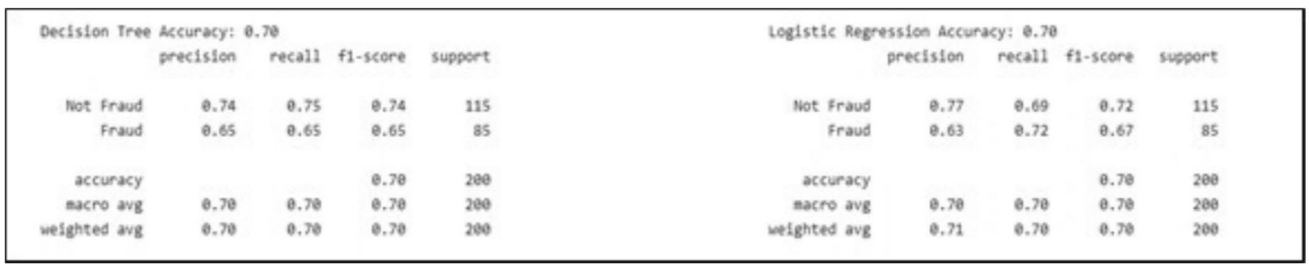
Fig. 11 LIME: Local interpretability prediction for fraud. From the figure, it is evident that the feature that most influences non-fraud transactions is the 'compliance flag,' while the feature that most influences fraud transactions is the 'anomaly flag.'

Table 8 Shapley Lorenz values for the top 10 features related to fraud transactions

S. no.	Feature name	Shapley Lorenz value
1	Both_Fraud_AML_Ind	0.3559
2	IsFraud	0.2847
3	Risk_Score	0.1423
4	Transaction_Anomaly_Score	0.0711
5	Transaction_Amount	0.0569
6	Transaction_Frequency_Indicator	0.0356
7	Anomaly_Flag	0.0213
8	Compliance_Flag	0.0142
9	Previous_Fraud_Reports	0.0107
10	Irregular_Behavior_Flag	0.0071

Some of the actionable insights that can be taken are:

- Transaction amount
  Action: AML Investigation for transactions with a transaction amount greater than a certain amount is immediately escalated to the next level of investigators for review.
  Benefit: Avoids fraud for a lump sum of money.
- Risk score.
  Action—continuously track the performance of the risk scoring engine. Set high priority for transactions with high risk score so that these are escalated for immediate attention.
  Benefit—Improves efficiency by triaging only the most critical alerts.
- Transaction Anomaly score
  Action—Adjust alert threshold logic for Transaction Anomaly score for transactions with a high score so that these are escalated for immediate attention.
  Benefit—Helps identify the pattern in anomaly scoring, thereby tuning the anomaly scoring rules.
- Compliance flag
  Action: check if the customer has all the KYC info in the system, and what is causing the compliance gap, and place the accounts on HOLD.
  Benefit: Avoids customers from committing further fraud.
- Transaction frequency indicator
  Action: Check the frequency of transactions and the aggregate value of money involved to understand the pattern of deposits. Include the findings in the report.
  Benefit: Update the rules to capture such transactions.



Decision Tree Accuracy: 0.70

	precision	recall	f1-score	support
Not Fraud	0.74	0.75	0.74	115
Fraud	0.65	0.65	0.65	85
accuracy			0.70	200
macro avg	0.70	0.70	0.70	200
weighted avg	0.70	0.70	0.70	200

Logistic Regression Accuracy: 0.70

	precision	recall	f1-score	support
Not Fraud	0.77	0.69	0.72	115
Fraud	0.63	0.72	0.67	85
accuracy			0.70	200
macro avg	0.70	0.70	0.70	200
weighted avg	0.71	0.70	0.70	200

Fig. 12 IE fraud prediction accuracy data. The statistics reveal that the accuracy of the decision tree and logistic regression are the same. However, the F1-score of fraud transactions is slightly higher in the decision tree compared to logistic regression

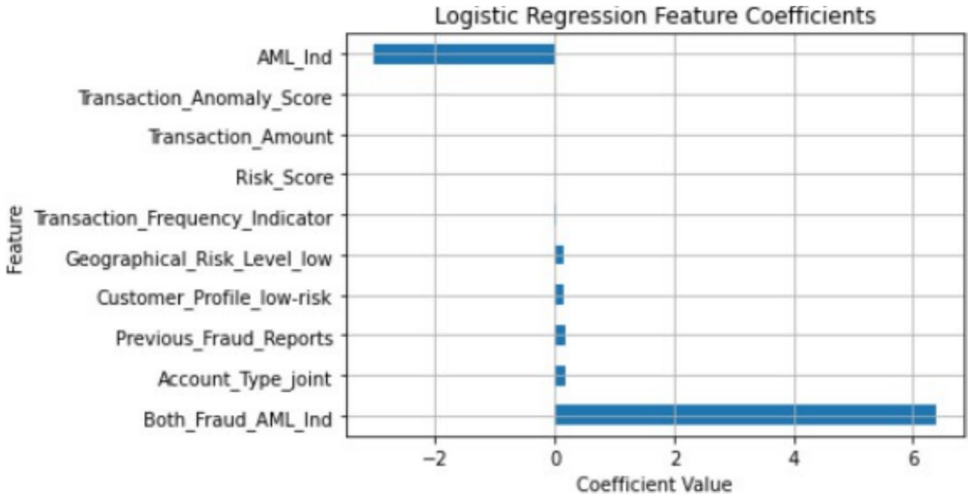


Fig. 13 Coefficient plot for fraud. XAI dashboard for Intrinsic explainer. From the coefficient plot, the most important feature in determining a fraud transaction is "Both fraud and AML indicator."

Table 9 Feature importance comparison from various models for fraud transactions

SHAP	LIME	Shapley Lorenz	Intrinsic explainer
Transaction amount	Compliance flag	Both fraud and AML Ind	AML Ind
Risk score	Transaction frequency indicator	IsFraud	Both fraud and AML Ind
Transaction Anomaly score	Risk score	Risk score	
Transaction frequency indicator	Transaction amount	Transaction Anomaly score	
Compliance flag		Transaction amount	
		Transaction frequency indicator	

To illustrate the explicability aspect of the proposed framework, we present a representative example of a transaction that was flagged as suspicious by the RF model. SHAP values revealed that the most influential features driving the prediction were a high risk score, an elevated transaction anomaly score, and a large transaction amount. These three variables had the highest positive SHAP values, pushing the model output decisively toward the fraud class. Additional contributing factors included an active anomaly flag and a record of previous fraud reports linked to the account.

The LIME explanation for this transaction showed consistent results. It identified the risk score, transaction frequency indicator, and irregular behaviour flag as the top contributors influencing the fraud prediction. Interestingly, the compliance flag slightly negatively affected the prediction, lowering the probability due to a recent verification step performed on the customer account.

The alignment between SHAP and LIME explanations reinforces the reliability of the model's decision and highlights the transparency of the framework. For investigators, such outputs provide interpretable, feature-level justifications for why a transaction is flagged, supporting both operational decision-making and compliance auditing.

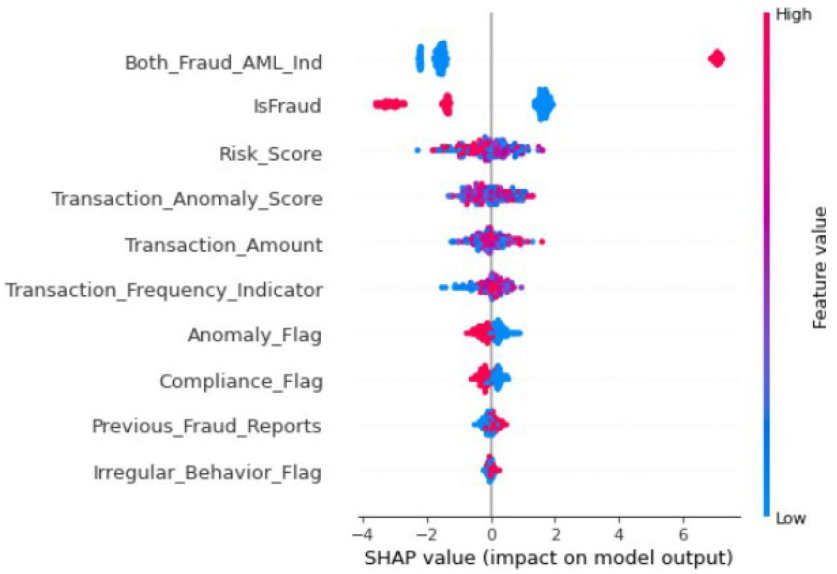
4.2 AML

SHAP technique Similar experiments performed for Fraud detection and interpretation were used for AML detection and interpretation, including feature optimisation. The model yielded a performance accuracy of 72%. The top 10 contributing features were plotted in the SHAP graphs for the SHAP summary plot, as seen in Fig. 14. The feature 'Both fraud and AML indicator' appears to be the top contributing feature in deciding whether the transaction made is an AML or not. The second significant contributing feature is the 'fraud indicator'. The top two contributing features imply that the majority of the transactions pertaining to AML are also linked to a fraud transaction. The third feature that significantly contributes is the 'risk score', which represents the risk assigned to a customer based on various factors such as geographical area, type of business, source of income, credit score, etc.

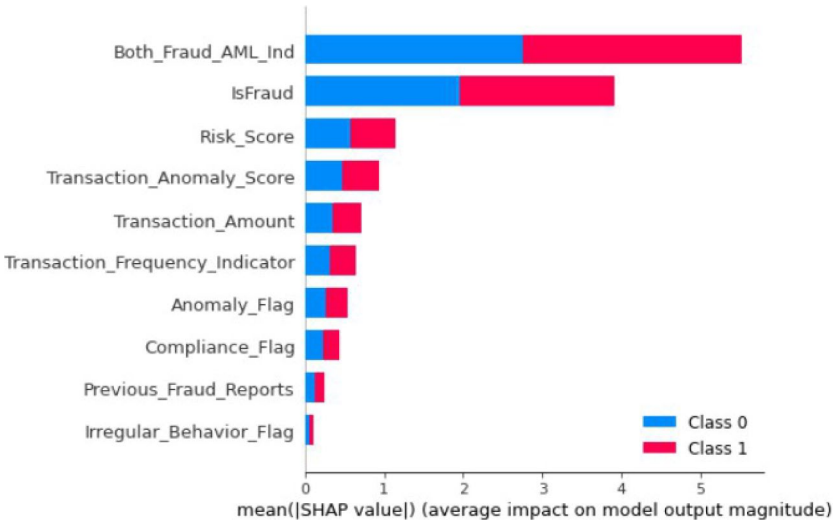
LIME Technique LIME explainer was applied to interpret the model outcome. The top two contributing features were identified to be the same as interpreted by the SHAP explainer, which is 'Both fraud and AML indicator' and the 'fraud indicator'. The third contributing feature is the 'Anomaly flag', meaning an outlier transaction that does not match the normal frequency of the transactions made. This can be seen in Figs. 15 and 16.

Shapley-Lorenz values Table 10 summarizes the findings of the calculation of Shapley-Lorenz values for AML.

The top 4 features—Both_Fraud_AML_Ind, IsFRaud, Risk_Score, and Transaction_Anomaly_score account for over 50% of the total model influence (~64%). This indicates a high concentration of predictive power in these features and a moderate inequality. The remaining features



(a) SHAP value for AML



(b) Mean SHAP value for AML

Fig. 14 SHAP plots for AML. The figure shows the SHAP plots for the top 10 contributing features of the model. The features 'Both fraud and AML indicator' and 'Is Fraud' are the model's top and least contributing features, respectively

contribute relatively less compared to the top 4 features. Similar to fraud, banks should consider innovating effective ways of risk rating the customers and improving the transaction anomaly scoring engines to reduce the impact of money launderers.

Intrinsic Explainer Similar experiments were conducted for fraud using intrinsic explainers for the AML data. This includes the feature optimisation method for considering the minimal features contributing to the overall model. A set of 10 contributing features were selected for interpreting the model. With the optimised features, the model was first trained using the decision tree. The accuracy was found to be 72%. With the same optimised features, the model was once again trained using logistic regression, and the accuracy was found to be 75%. In this case, the accuracy was slightly higher for logistic regression compared to the decision tree classifier, as seen in Fig. 17.

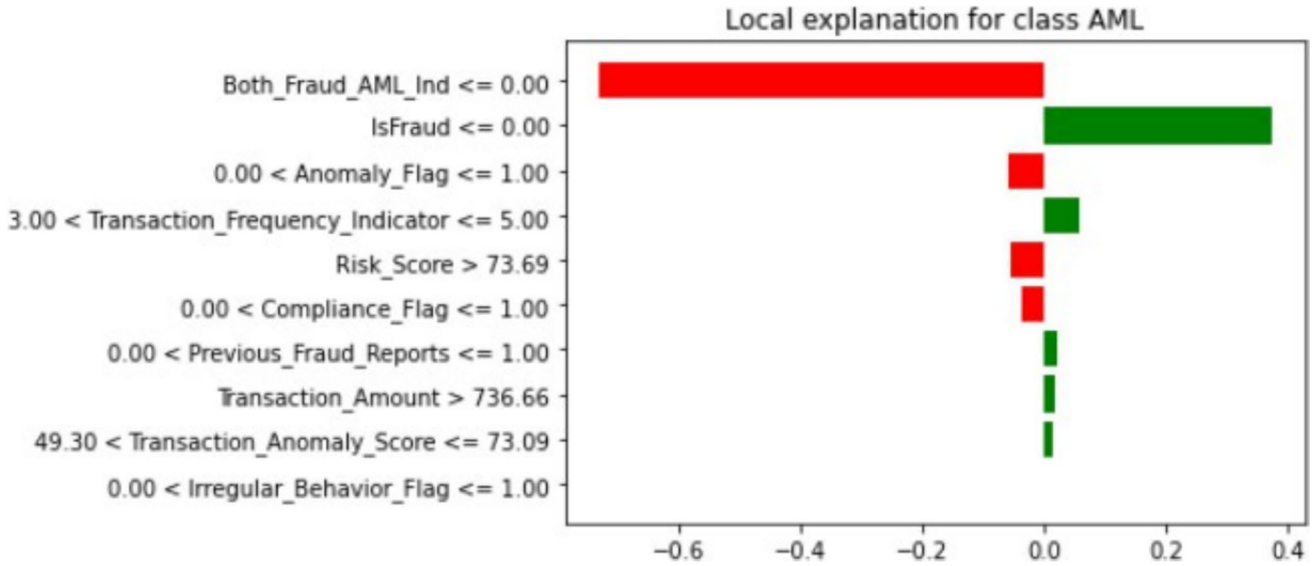


Fig. 15 LIME: feature importance for AML. The figure shows the LIME plot for the top 10 contributing features of the model built. The features 'Both fraud and AML indicator' and 'Is Fraud' are the top and the least contributing features of the model, respectively

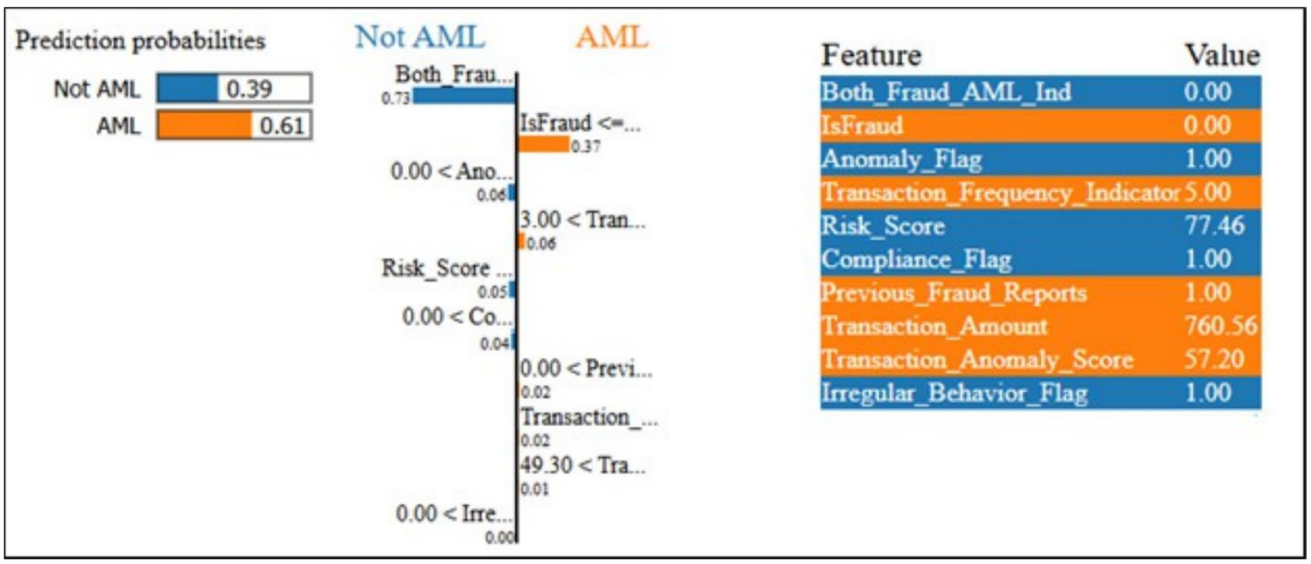


Fig. 16 LIME: local interpretability of AML. From the figure, it is evident that the feature that most influences non-AML transactions is the 'Both fraud and AML indicator,' while the feature that most influences AML transactions is the 'Is Fraud.' The plot also shows the confidence interval, stating that this data consists of 39% AML and 61% non-AML transactions, respectively

Table 10 Shapley Lorenz values for the top 10 features related to AML transactions

S. no.	Feature name	Shapley Lorenz value
1	Both_Fraud_AML_Ind	0.20
2	IsFraud	0.18
3	Risk_Score	0.14
4	Transaction_Anomaly_Score	0.12
5	Transaction_Amount	0.11
6	Transaction_Frequency_Indicator	0.09
7	Anomaly_Flag	0.06
8	Compliance_Flag	0.04
9	Previous_Fraud_Reports	0.03
10	Irregular_Behavior_Flag	0.03

A coefficient plot was generated for the logistic regression classifier to study the correlation. From Fig. 18, it was identified that the 'Fraud indicator' and the 'Both Fraud and AML' indicators are strongly correlated, while the other features are weakly correlated. This also highlights a strong correlation between fraud and AML records. The other significant feature that contributes is the stage at which the AML transaction was spotted, and in this case, it was found to be the staging area ('AML_Stage_Ind_S').

Comparison A comparison of the outcome of feature importance has been studied for AML models, as seen

Decision Tree Accuracy: 0.72

	precision	recall	f1-score	support
Not AML	0.73	0.75	0.74	106
AML	0.71	0.68	0.70	94
accuracy			0.72	200
macro avg	0.72	0.72	0.72	200
weighted avg	0.72	0.72	0.72	200

Logistic Regression Accuracy: 0.75

	precision	recall	f1-score	support
Not AML	0.76	0.76	0.76	106
AML	0.73	0.73	0.73	94
accuracy			0.75	200
macro avg	0.75	0.75	0.75	200
weighted avg	0.75	0.75	0.75	200

Fig. 17 Coefficient plot for AML. The statistics reveal that the accuracy of the logistic regression is higher than decision tree. The accuracy of logistic regression is 75%. The other stats such as precision and recall were also noted to be higher in logistic regression

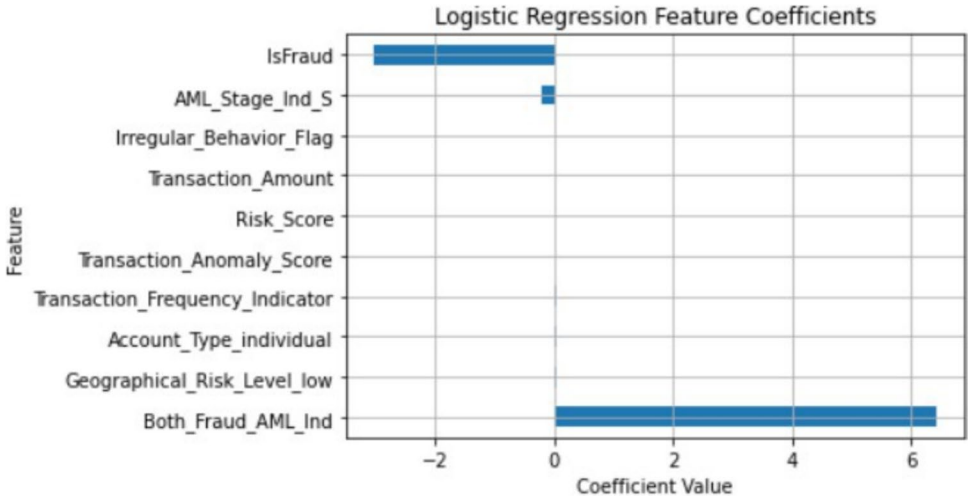


Fig. 18 Coefficient plot for AML. AML XAI dashboard for Intrinsic explainer. From the coefficient plot, the most important feature in determining an AML transaction is "Both fraud and AML indicator."

Table 11 Feature importance comparison from various models for AML transactions

SHAP	LIME	Shapley Lorenz	Intrinsic explainer
Both fraud and AML Ind	IsFraud	Both fraud and AML Ind	IsFraud
IsFraud	Transaction Anomaly score	IsFraud	Both fraud and AML Ind
Risk score		Risk score	
Transaction Anomaly score		Transaction Anomaly score	
Transaction frequency indicator		Transaction amount	
		Transaction frequency indicator	

in Table 11. The most consistent features across all four explainers are 'Is Fraud indicator', 'Transaction Anomaly score', 'Risk score', and 'Transaction frequency indicator'.

Some of the actionable insights that can be taken are:

- Isfraud
  Action: Check if the transaction is also linked to a fraud transaction in addition to the AML transaction.
  Benefit: The appropriate regulatory authority can be notified for legal action.
- Risk score
  Action—Continuously track the performance of the risk scoring engine. Set high priority for transactions with high risk score so that these are escalated for immediate attention.
  Benefit—Improves efficiency by triaging only the most critical alerts.
- Transaction Anomaly score
  Action—Adjust alert threshold logic for Transaction Anomaly score for transactions with a high score to escalate these for immediate attention.
  Benefit—Helps identify the pattern in anomaly scoring, thereby tuning the anomaly scoring rules.
- Transaction frequency indicator
  Action: Check the frequency of transactions made and analyze what stage the transaction is made at, such as staging, layering, etc. Include the findings in the report.
  Benefit: Update the rules to capture such transactions at early laundering stages.
- Anomaly flag
  Action: Investigate the anomaly causing the rule and act accordingly.
  Benefit: Continuous monitoring of the algorithms will help improve the business rules.

Like fraud detection, explainability was also applied to a representative AML case flagged by the model. This transaction involved a series of medium-sized transfers from a corporate account with recent irregular activity. SHAP analysis identified the top contributing features as a high risk score, elevated transaction anomaly score, and frequent transactions as indicated by the transaction frequency indicator. These features had the most substantial positive impact on the AML classification. The compliance flag reflected incomplete documentation, and the irregular behaviour flag further reinforced the alert.

LIME results were consistent, highlighting risk score, transaction anomaly score, and previous fraud reports (which increased the account's risk profile) as major contributors. The LIME local explanation also revealed that if the transaction had occurred in a lower-risk jurisdiction or had a lower frequency score, the alert probability would have been significantly reduced.

This example demonstrates how post-hoc explainability methods can help deconstruct and justify AML risk predictions at a granular level. By pinpointing the most influential factors, the system supports compliance teams in quickly validating alerts and taking appropriate action in line with AML regulatory requirements.

4.3 AML and Fraud

SHAP Similar experiments were performed for fraud, and AML detection and interpretation were used to detect and interpret a transaction that is both AML and fraud. The model yielded a performance accuracy of 100%. The top 10 contributing features were plotted in the SHAP graphs for the SHAP summary plot as seen in Fig. 19. The features 'fraud indicator' and 'AML indicator' appear to be the top two contributing features in deciding whether the transaction made is both 'AML and fraud'.

LIME Technique LIME explainer was applied to interpret the model outcome. The top two contributing features were identified to be the same as interpreted by the SHAP explainer: The 'AML indicator and the fraud indicator'. Other features that contributed to the model selection can be seen in Figs. 20 and 21, but the contribution was not so significant. E.g., risk score, AML stage indicator, transaction anomaly score, etc.

Shapley-Lorenz values Table 12 summarizes the findings of the calculation of Shapley-Lorenz values for AML and fraud.

The top 2 features—IsFraud and AML_Ind account for ~ 74 of the model's predictive influence. The remaining eight features collectively contribute just ~ 26%, indicating a strong feature dominance. The Gini coefficient is calculated and found to be ~ 0.63, which suggests a heavily imbalanced feature contribution, and this is a typical pattern in fraud and AML systems where a few flags dominate. It can be inferred that there are high chances of a fraud transaction being a money laundering transaction as well, and vice versa. Banks should take measures and build frameworks to verify a confirmed fraud transaction, whether it is a money laundering transaction as well. In addition to these steps, the bank should have some checks on the risk rating of the customer and the value of the transaction amount.

Intrinsic Explainer Similar experiments conducted in the above sections were performed to interpret 'both AML and fraud' data. This includes the feature optimisation method used to consider the minimal features that are contributing to the overall model. A set of 10 contributing features was selected for interpreting the model. With the optimised features, the model was first trained using the decision tree. The accuracy was found to be 100%. With the same optimised features, the model was once again trained using logistic regression, and the accuracy was found to be 100%, as seen in Fig. 22. In this case, the accuracy result was the same for both logistic regression and the decision tree classifiers.

A coefficient plot was generated for the logistic regression classifier to study the correlation, as seen in Fig. 23. From the plot, it was identified that the 'Fraud indicator' and the 'AML indicator' were strongly correlated. Other fraud-related features, such as fraud-type MIP (malicious IP) and PP (phone porting), were also correlated. AML-related features, such as AML stage indicators (placement and staging), were also correlated and contributed to the model.

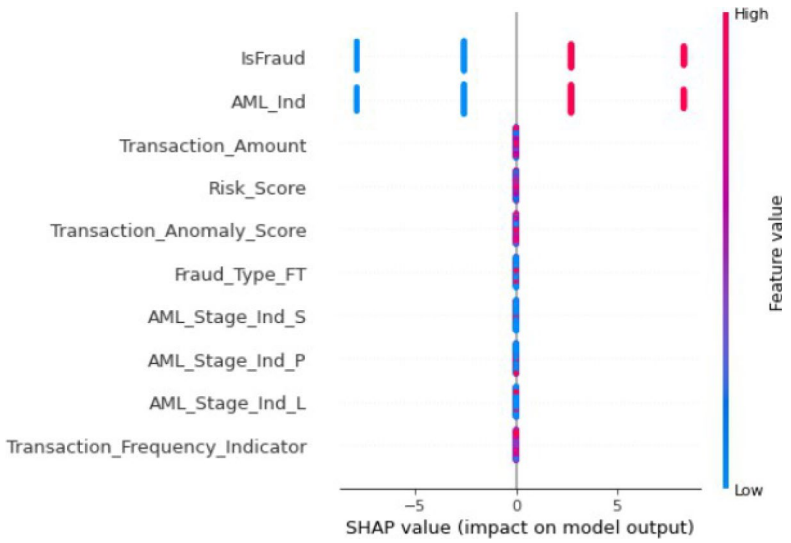
The study revealed the top 10 contributing features to fraud, AML, and 'fraud and AML' transactions. The coefficients of the features exhibiting strong correlations were also identified. One of the findings is that there is a strong correlation between fraud and AML transactions.

Comparison

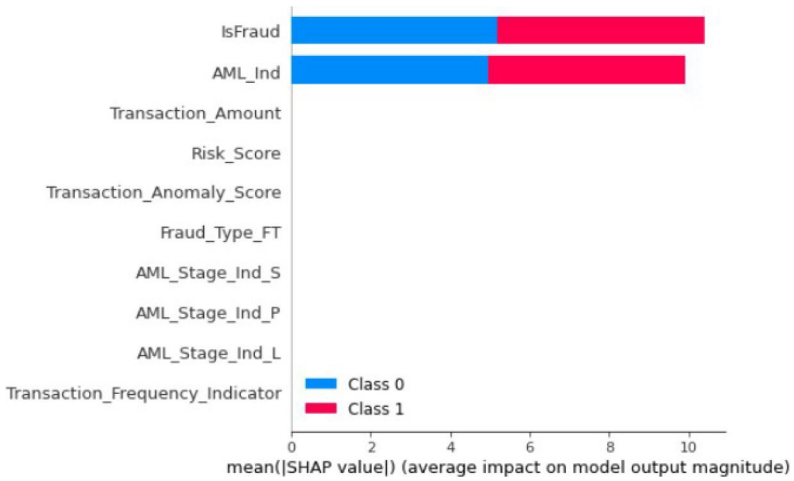
A comparison of the outcome of feature importance has been studied for AML and fraud models, as seen in the Table 13. All four explainers unanimously identify 'Is Fraud' and "AML indicator' as key features highlighting the strength of these indicators as pivotal decision drivers within models targeting financial crime detection.

Some of the actionable insights that can be taken are:

- Isfraud
  Action: Check if the transaction is also linked to a fraud transaction in addition to the AML transaction.
  Benefit: The appropriate regulatory authority can be notified for legal action.
- AML_Ind
  Action: Investigate the laundering stages where transactions are made. Update the rules accordingly.



(a) SHAP value for AML and Fraud



(b) Mean SHAP value for AML and Fraud

Fig. 19 SHAP plots for AML and fraud. The figure shows the SHAP plots for the top 10 contributing features of the model. The features 'Is fraud' and 'Transaction frequency indicator' are the model's top and least contributing features, respectively

Benefit: prospect laundering cases are likely to reduce, leading to adherence to compliance and avoidance of loss of money.

5 Conclusions and future works

This study proposed XAI models for detecting suspicious banking transactions, including fraud and AML activity. Our experimental results show that GB achieved the highest F1-score (0.78) in fraud detection, while RF performed best in AML detection (F1-score: 0.63). SVM also yielded competitive results, particularly in recall-sensitive scenarios. Furthermore, statistical significance testing and comparison analysis confirmed that the suggested models perform better than baseline classifiers and are of good reliability. The strength of these results gives us further confidence that our methodology can detect suspicious transactions in the real world with reliable, regulator-friendly performance. Regarding explainability, both SHAP and LIME methods successfully identified key contributing features—such as risk score, transaction anomaly score, previous fraud reports, and irregular behaviour flags—which provided actionable

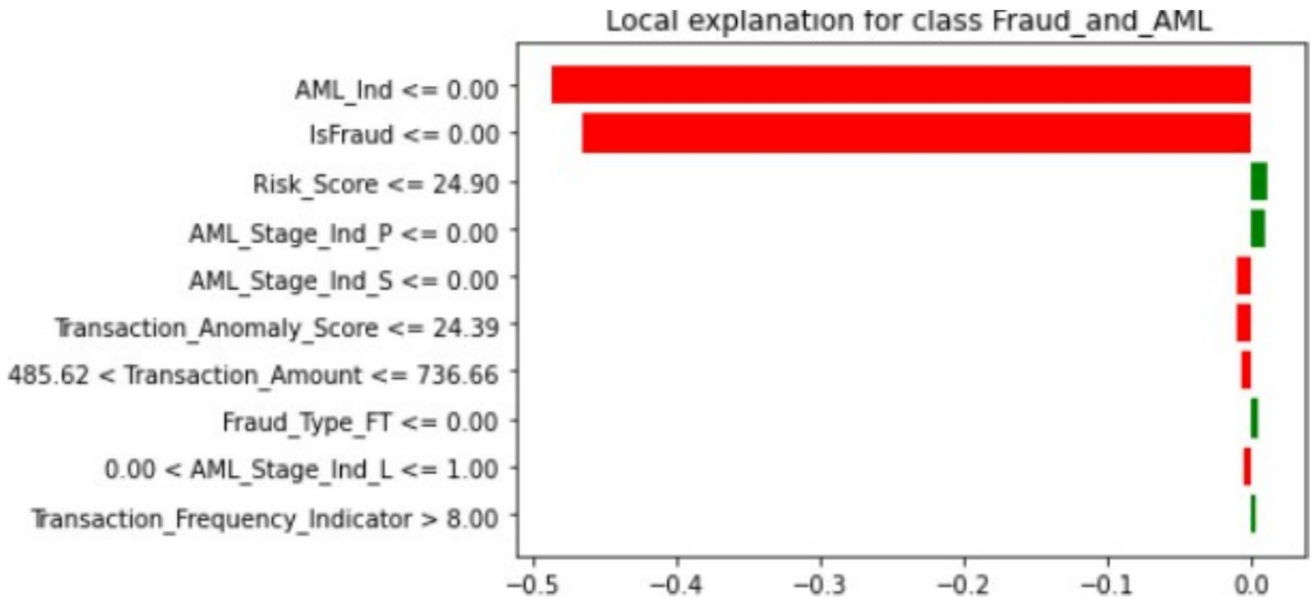


Fig. 20 LIME: Feature importance for fraud and AML. The figure shows the LIME plot for the top 10 contributing features of the model built. The features 'AML indicator' and 'Transaction frequency indicator' are the top and the least contributing features of the model, respectively

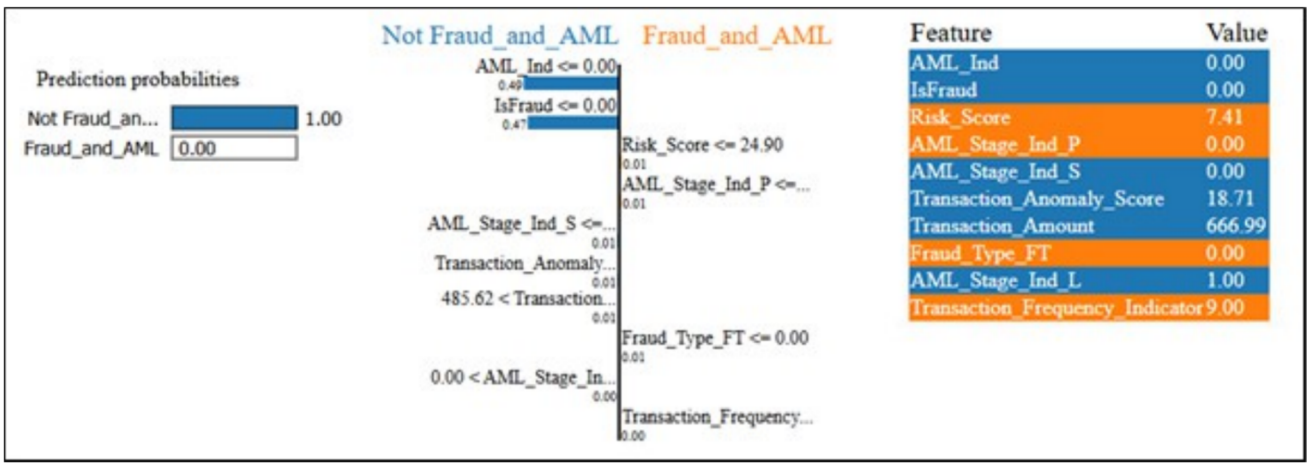


Fig. 21 LIME: Local interpretability of fraud and AML. From the figure, it is evident that the feature that most influences 'non-AML and non-fraud' transactions is the 'AML indicator,' while the feature that most influences 'AML and fraud' transactions is the 'Risk score.' The plot also shows the confidence interval, stating that this data consists of 100% 'AML and fraud' and 0% 'non-AML and non-fraud' transactions, respectively

Table 12 Shapley Lorenz values for the top 10 features related to 'AML and fraud transactions'

Sl. No	Feature Name	Shapley Lorenz Value
1	IsFraud	0.3846
2	AML_Ind	0.3526
3	Transaction_Amount	0.0962
4	Risk_Score	0.0513
5	Transaction_Anomaly_Score	0.0321
6	Fraud_Type_FT	0.025
7	AML_Stage_Ind_S	0.0192
8	AML_Stage_Ind_P	0.0192
9	AML_Stage_Ind_L	0.0128
10	Transaction_Frequency_Indicator	0.0064

insights into why certain transactions were flagged. These findings validate the potential of XAI-enhanced models in supporting compliance analysts, improving the transparency of decision-making, and facilitating auditability in regulated banking environments. Our inclusion of Shapley-Lorenz values introduces a normalized, inequality-aware approach to feature attribution, enabling clearer global comparisons and fairer decision support. These contributions offer value to model developers and compliance teams and regulators who require interpretable and auditable outputs from complex machine learning systems. One of the discoveries was that there is a strong correlation between fraud and AML records, which was revealed when 'fraud

Coefficient Value

Decision Tree Accuracy: 1.00

	precision	recall	f1-score	support
Not Fincrime	1.00	1.00	1.00	162
Fincrime	1.00	1.00	1.00	38
accuracy			1.00	200
macro avg	1.00	1.00	1.00	200
weighted avg	1.00	1.00	1.00	200

Logistic Regression Accuracy: 1.00

	precision	recall	f1-score	support
Not Fincrime	1.00	1.00	1.00	162
Fincrime	1.00	1.00	1.00	38
accuracy			1.00	200
macro avg	1.00	1.00	1.00	200
weighted avg	1.00	1.00	1.00	200

Fig. 22 IE AML and fraud prediction accuracy analysis. The statistics reveal that the accuracy of the logistic regression is same as decision tree

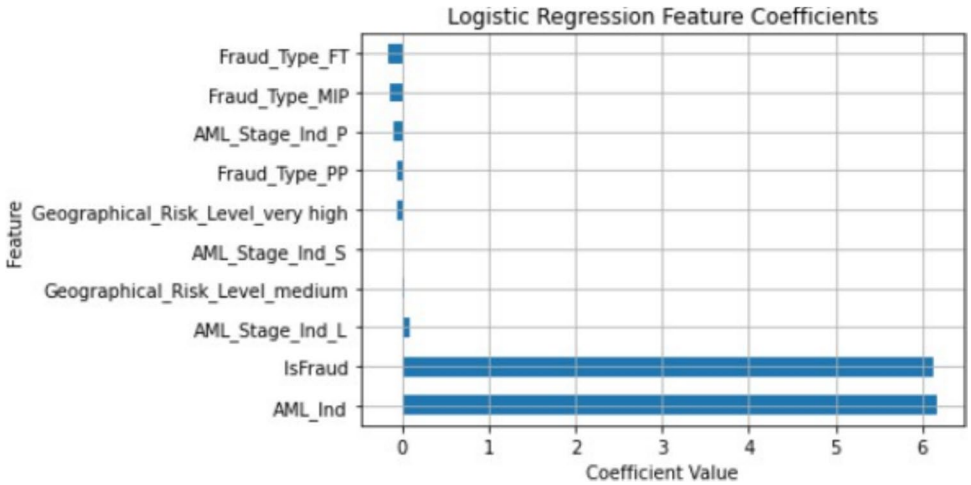


Fig. 23 Coefficient plot for AML and fraud: 'Fraud and AML XAI' dashboard for Intrinsic explainer. From the coefficient plot, the most important feature in determining a 'Fraud and AML transaction' is "AML indicator."

Table 13 Feature importance comparison from various models for AML transactions

SHAP	LIME	Shapley Lorenz	Intrinsic Explainer
IsFraud	IsFraud	IsFraud	IsFraud
AML Ind	AML Ind	AML Ind	AML Ind
		Transaction amount	
		Risk score	
		Transaction Anomaly score	

and AML' transactions were analysed together. From a statistical viewpoint, it was noticed that the feature optimized models balance accuracy and interpretability. Strong results on combined AML and fraud suggest good signal strength, while low accuracy of fraud detection signals the need to further handle the data. RFE was used effectively to reduce complexity while keeping interpretability. The model was constrained to the top 10 features, maintaining explanatory power.

This research aligns strongly with the Safe Machine Learning paradigm, emphasizing transparency, robustness, and responsible decision-making in financial crime detection systems. By integrating feature selection (RFE) with XAI techniques—namely, SHAP, LIME, and intrinsic interpretable models—the proposed framework not only enhances model efficiency but also improves interpretability. The ability to identify and validate the top contributing features across multiple explanation methods ensures trustworthiness and accountability, addressing both regulatory requirements and investigator needs. The use of coefficient plots and SHAP summaries bridges the gap between statistical performance and explainable outcomes, making the system suitable for high-stakes domains such as AML and fraud detection. These qualities directly support the principles of Safe machine learning, including fairness (bias awareness), transparency (clear reasoning), robustness (feature optimization), and human oversight (interpretation by compliance officers and auditors) [120].

Further study can be considered to address the model's approach to treating false positives and identify if there are any instances where XAI fails to provide explanations. Two of them are:

Model Debugging: Use XAI for Feedback Loops

- Use explanations of false positives to:
- Re-label or augment training data.
- Tune thresholds or retrain models.
- Design rules-based overrides (e.g., whitelist known behavior patterns).

Practical Tools and Practices

- Use counterfactual explanations (e.g., "what minimal change would make this not a fraud case?") to understand decision boundaries.
- Consider visual dashboards integrating SHAP/LIME for investigators.

Future research should also explore advancing financial crime detection, which requires a balanced approach that combines the predictive strength of deep learning with the interpretability of traditional machine learning, incorporating real-time transaction scoring, and evaluating cross-institution generalisability. Additionally, embedding the XAI explanations into human-in-the-loop workflows for compliance teams may further enhance interpretability and practical adoption in live systems. Deep learning models excel at capturing intricate, non-linear patterns in transactional data, making them well-suited for uncovering sophisticated or evolving criminal tactics. However, their lack of transparency poses significant challenges in regulatory contexts. Integrating XAI methods—such as SHAP and LIME—alongside intrinsically interpretable models like decision trees or generalized additive models can help bridge this gap. This hybrid approach enhances detection performance and supports accountability and trust in automated decision-making. Moving forward, such integrative frameworks are essential for building robust, transparent, and compliant financial crime detection systems.

Currently, the governments have different frameworks and regulators dealing with fraud and AML separately. There is no common framework to deal with both. This is a gap in the regulation process. The information is not shared between the regulators at the same time. A combined framework proposal to the regulators could be interesting in the future. An integrated AML and fraud risk management platform with a shared data foundation, joint risk scoring, unified case and reporting workflows, and consistent XAI tooling is recommended. This approach enhances detection, reduces false positives, satisfies AUSTRAC and AFCX obligations, and prepares for convergence-focused regulatory expectations.

Acknowledgements Not applicable.

Authors' contributions All authors contributed to the study's conception and design. Material preparation by Narasimha Kumar Narasapuram, under the supervision of other authors, data collection and analysis were performed by Narasimha Kumar Narasapuram. The first draft of the manuscript was written by Narasimha Kumar Narasapuram and all authors commented on previous versions of the manuscript. All authors read and approved the final manuscript.

Funding Open Access funding enabled and organized by CAUL and its Member Institutions

Data availability The dataset and the related code used in this study will be made publicly available upon acceptance of the manuscript.

Declarations

Conflict of interest The authors declare no competing interests.

Ethical approval Not applicable.

Consent to participate Not applicable.

Consent for publication Not applicable.

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